SECTION 909

TREATED WOOD FENCE PRODUCTS

909.01 Scope – This Section covers the quality and manufacturing requirements for preservative Pressure Treated Wood Fence Products ("Fence Products") such as fence posts, gateposts, braces, and droppers used in Ministry Work.

Treated wood shall meet the requirements of this SS 909 as well as the product-specific requirements in the following related Standard Specification sections:

(a) SS 700 – Wildlife Exclusion Fencing

(b) SS 741 – Fence Construction

(c) SS 908 – Preservative Treatment – Wood Products

909.02 General – Fence Products shall be supplied in the sizes, species, and grading specified, in conformity with this Section, and as generally shown on the applicable drawings in SS 700 and SS 741.

Fence Products shall be newly manufactured products. Previously used Fence Products are not accepted for the Work.

909.03 Material – Wood Species – All Fence Products such as round fence posts (including line, straining, corner, intersection, gate and end posts) and any required round wood braces and droppers shall be Lodgepole Pine or Jack Pine cut from live growing trees.

Use of other species or use of wood not cut from live trees (such as from slash piles, blown down trees, or kill stands) must be as specified in the Contract, Drawings or Special Provisions or must be approved in writing by the Ministry Representative prior to supply. Cedar, either split or whole, will not be approved.

Tree species not meeting specified requirements are subject to rejection with no payment to the Supplier and/or Contractor.

909.04 Preservation Preparation and Wood Quality – All fence posts and bracing shall be clean peeled with minimum removal of sapwood for their full length with all bark and cambium removed, and free from glazed surfacing left by dried sap.

All materials shall have sufficient sapwood in the outer periphery to obtain the specified penetration of preservative.

All sound knots or projections shall be shaved smooth and flush with the surface of surrounding wood.

909.05 Defects

909.05.01 Prohibited Defects – All round fence posts and braces shall be free of the following defects:

(a) Decay, rot, stain and red heart;

(b) Holes (except for test purposes, which shall be properly plugged);

(c) Nails, spikes and other metal or foreign substances;

(d) Loose or unsound knots including spike knots and knot clusters;

(e) Excessive sweep, reverse sweep and short crook;

(f) Cracks, splits or through checks, other than limited defects as defined below;

(g) Shakes in the top or butt, and

(h) Unsound scars (cat faces).

909.05.02 Limited Defects – All round fence posts and braces are permitted limited, in the opinion of the Ministry Representative, defects as follows:

(a) Season checks less than 10 mm in width if less than 500 mm in length, or less than 5 mm in width if more than 500 mm in length.

(b) The depth of splits or checks shall not exceed the depth of 40% of the post diameter or thickness, regardless of their length or width.

(c) Sound scars (cat faces) permitted if the depth of the trimmed scar is less than 20 mm in depth, less than 300 mm in length, and which does not impair strength and durability

(d) Sound, tight, well-spaced knots permitted provided they do not exceed 38 mm in diameter

(e) Insect damage consisting of holes 1.6 mm or less in diameter. Scoring or channelling of the surface is permitted. All other forms of insect damage are prohibited.

(f) Two coats of field treatment preservative shall be applied to splits and checks that meet the criteria for limited defects and that occur after pressure treatment.

909.06 Treated Material Requirements

909.06.01 General – Round wood posts, sawn timber posts and other wood fence components shall be treated in accordance with SS 908 and this Section.

Where applicable, treatment to "refusal" shall be as defined in CSA O80:

> ***Treatment to refusal (refusal treatment)*** *— treatment of wood under specified conditions until the quantity of preservative absorbed during two consecutive half-hours is not more than 2% of the amount already injected.*

909.06.02 Round Fence Posts and Braces Dimensions – All dimensions shall apply inside the bark and to the fully seasoned and treated fence posts or braces.

(a) Dimensions

(i) Diameter Class – Round fence posts and braces shall be classified as to size on the basis of the smaller diameter and length.

(ii) Length and Diameter – Wood post length and diameter shall be as specified SS 700 and SS 741 and their respective SP Drawings or as specified in the Contract.

The diameter specified shall be the minimum with a tolerance of +20 mm and the length shall not vary by more than ±25 mm from that specified.

(iii) Taper – The maximum permitted taper shall be such that the larger diameter shall not exceed that of the specified diameter by more than 25 mm over a 2 m length of post or brace.

The maximum permitted taper shall be proportional for posts and braces of lengths other than 2 m.

In cases where the post or brace is not truly circular, the diameter shall be obtained by measuring the circumference and dividing by 3.14.

(iv) Straightness (Sweep) – When dried, all posts shall be:

(A) straight;

(B) free from sweep, short crooks or reverse sweep; and

(C) free of bends in more than one plane.

Straightness shall be measured by a straight line joining the mid-point of the butt and the mid-point of the tip.

The straight line shall not fall outside the body of the post, nor be more than 50 mm from the geometric centre of the post at any point.

(b) Post Ends – Ends of posts and braces shall be cut square to the specified length and, unless otherwise specified, the butt end (larger diameter) shall be machine "pencil" pointed or chiselled before treatment to permit driving of posts.

The diameter at the point shall be not less than 18 mm and not more than 35 mm, and the pointing shall extend over a length of 150 mm to 200 mm of the post, or opposite sides shall be tapered to a "chisel point" of similar basic dimensions.

The top (smaller diameter) end of the post shall be domed (chamfered) or rounded 15 mm to 30 mm at approximately 30º to 45º around the circumference edge.

909.06.03 Sawn Lumber Posts and Braces – Gate posts and braces not stipulated as round, together with any anchor cleats, shall be Standard and Better S4S Lodgepole Pine, Jack Pine or Coast Douglas Fir, to the current NLGA grading rules and in the required lengths and sizes.

909.06.04 Fence Droppers Dimensions – Wood droppers to stabilize barbed wire fencing (Type C) between posts shall be one of the following:

(a) Round fence posts Lodgepole Pine, Jack Pine of 50 mm (smaller) diameter, or

(b) 25 mm x 50 mm sawn lumber, Standard and better, S4S, Lodgepole Pine, Jack Pine or Coast Douglas Fir, to the current NLGA grading rules,

(c) 50 mm x 50 mm sawn lumber, Standard and better, S4S, Lodgepole Pine, Jack Pine or Coast Douglas Fir, to the current NLGA grading rules.

Wood dropper length shall be as specified SS 700 and SS 741 and their respective SP Drawings or as specified in the Contract.

909.07 Pressure Treatment

909.07.01 Sampling Zone – The sample zone for penetration and retention shall be to at least a depth of 25 mm from the surface.

909.07.02 Preservative Penetration and Retention – Retention and penetration shall be in accordance with SS 908, as per CSA O80 UC4.1.

For sawn or dressed fence braces, anchor cleats and droppers, the Ministry Representative may accept a preservative penetration depth "to refusal" upon review of the plant records of penetration time applied for these types of Fence Products.

The treatment plant shall be responsible to certify and document that the final Fence Products meet the minimum retention and penetration requirements.

Materials not meeting the minimum specified penetration depths may be rejected by the Ministry Representative at the Supplier's or Contractor's expense.

909.08 Quality Assurance – The Ministry Representative may conduct, or may have a third-party conduct, quality assurance testing and reviews.

SECTION 911

STEEL AND IRON

911.01 Scope – This standard only applies to materials used in the maintenance and rehabilitation of existing log and timber bridges and log or timber components such as log crib retaining walls, timber piers, wood sign structures, etc.

911.02 Materials – These materials shall be in accordance with the current editions of the specifications indicated in Table 911-A.

Table 911-A: Iron and Steel Materials Specifications

Material	Specification
Structural Steel	CSA G40.21 Grade 260 or better
Steel Bolts	ASTM A307
Steel Nuts	ASTM A563
Carbon Steel Castings	ASTM A27
Iron Castings	ASTM A48
Pipe Steel	ASTM A53

911.03 Tension and Lateral Rods – Tension and lateral rods shall be of structural steel.

(a) Upsets Ends – Upset ends shall be upset by hand or machine and welds will not be allowed. The dimensions of upset ends shall be made as shown on the Drawings. When upsetting by machine, the ends shall be upset to a little oversize, after which they should be heated to welding temperature and then swaged to the proper size.

(b) Nuts – The nuts shall be made hexagonal, U.S. Standard sizes and threads.

(c) Threads – The threads on rods and nuts shall be full, smooth, uniform and of the same pitch throughout. They shall be such that the nuts can be run on by hand the full length of the threads on the rods without showing undue or uneven slackness.

911.04 Machine Bolts – Machine bolts shall conform to ASTM A307.

911.05 Carriage Bolts – Carriage bolts shall conform to ASTM A307. They shall be of the common type with button head, square neck and square or hex nut.

911.06 Drift Bolts or Pins – Drift bolts or pins shall be of structural steel. They shall be cut from plain, round bars unless otherwise called for on the Drawings. The ends of drift pins or bolts shall be tapered or shaped so that the pin may be easily driven into pre-bored holes in the wood being fastened.

911.07 Dowels – Dowels shall be of structural steel. They shall be cut from plain round bars and any ragged ends shall be removed.

911.08 Lag Screws – Lag screws shall conform to ASME B18.2.1, be made of steel that meets or exceeds the properties of ASTM A307 Grade A or SAE J429 Grade 1, and have square heads and cone points.

911.09 Plate Washers or Wrought Washers – Plate washers or wrought washers shall be of structural steel or wrought iron; they shall be round or square as specified. Round washers shall be according to Canadian Manufacturer's Standards. Square washers shall be made to the dimensions called for on the Drawings.

911.10 Ogee Washers – Ogee washers shall be of cast iron. The proportions of these washers adopted by some of the manufacturers differ somewhat in detail, but to be acceptable the diameter should not be less than four times the diameter of the bolt for which it is to be used and the thickness should be approximately equal to the diameter of the bolt.

911.11 Special Castings – Special castings shall be true to pattern, free from excessive shrinkage or overrun, be made in accordance with good work practice, and be free from defects. Castings, which show defects after machining, will be rejected notwithstanding any previous acceptance at the manufacturer's works.

911.12 Steel Splice-Joint Fastenings – Plates and bars, etc., for these joints shall be of structural steel. All parts shall be made in accordance with the Drawings. Bolt holes shall be drilled from the solid or sub-punched and reamed. In sub-punched and reamed work, the holes shall be punched 3/16" smaller and after assembling, reamed 1/16" larger than the nominal diameter of the bolt.

911.13 Galvanizing – Unless otherwise specified, all steel hardware, steel fasteners and other steelwork shall be galvanized. Galvanizing shall be in accordance with ASTM A123 or ASTM A153, as applicable.

911.14 Plates – The various plates designated on the Drawings as gib, bearing, bed, joint, lateral, etc., shall be of structural steel.

911.15 Welding – Welding shall conform to the requirements of CSA W59.

911.16 Pipe Fastenings – Pipe fastenings shall be Schedule 80 pipe. Nuts shall be hexagonal machine lock nuts, U.S. Standard as to outside dimensions and threads. The threads on nuts and pipes shall be full, smooth, uniform

and of the same pitch; they shall be such that they can be run on by hand the full length of the thread on pipe without undue or uneven slackness.

911.17 Cleaning and Painting – Painting of steel and iron components shall be in accordance with SS 216.10 Application Methods (Excluding metalizing and Hot Dip galvanizing). No paint shall be applied to surfaces which are to be encased in concrete.

911.18 Machine Finished Surfaces – Machine finished surfaces and threaded ends of tension and lateral rods shall be coated with an anti-corrosion lubricant applied as soon as the surfaces and threads are finished and accepted by the Inspector.

911.19 Inspection – All materials and quality of work shall be subject to inspection by the Ministry Representative.

(a) The Contractor shall allow the Ministry Representative free access to the shops at all times when work on the materials is being done and the Contractor shall provide every reasonable facility to assist the Ministry Representative in the inspection of both material and quality of work.

(b) The Ministry Representative may reject material which does not meet the requirements of the Contract. Rejected material or poor quality of work shall be replaced promptly with materials meeting the requirements and quality at no cost to the Ministry.

911.20 Shipping – The threaded ends of all rods shall be protected from damage.

SECTION 941

PRECAST REINFORCED CONCRETE BARRIERS

941.01 Scope – This Section covers the quality and manufacture of precast reinforced concrete roadside and median barriers for highway and other off-highway traffic confinement use.

The concrete traffic barrier units shall be supplied in the sizes and types as required by the Contract, Purchase Order, Work Order or Drawings in strict conformity with this Section and pertinent SS Drawings of the SP941 Series.

941.02 Concrete

941.02.01 Concrete Quality

(a) Concrete quality shall conform to CSA A23.1 and CSA A23.2 except where amended hereafter. SS 211 shall only apply when specifically referenced.

(b) The manufacturer shall implement a quality control program to meet the specified requirements. The quality control plan shall be made available to the Ministry Representative for review.

(c) The Ministry may implement a quality assurance program by auditing the manufacturer's quality control program and by inspection and testing at its discretion.

(d) When requested by the Ministry Representative, the manufacturer shall submit concrete mix design information, using the requirements of CSA A23.1 Table 5, Alternative 1, for Ministry quality assurance and compliance review.

(e) A compressive strength test result is defined as the average of the strengths of three 28 day compressive test cylinder breaks.

(f) The strength level of the concrete represented by the test shall be considered satisfactory if the test result equals or exceeds 32 MPa and no individual cylinder strength is less than 28.5 MPa. If this condition is not met, the concrete will be considered to have failed the strength requirements. No other form of testing to prove the relative strength at a later date will be allowed without the approval of the Ministry Representative.

(g) Concrete testing cylinders shall be cast by the precast concrete manufacturer or their authorized representative at the time of placing concrete. Frequency of testing shall be in accordance with CSA A23.4 Clause 19. For compressive strength testing, a minimum of two cylinders shall be tested from each type of concrete mix each day or a minimum of two cylinders for each 30 m^3 of any one mix when daily production exceeds 30 m^3. When the daily quantities are small, one compressive strength test shall be performed for every 3 m3. When the daily quantities are smaller than 3 m3, two compressive strength tests shall be performed each week. Concrete cylinder samples may be collected by the Ministry Representative at any time and tested to ensure the concrete is meeting specification requirements.

(h) Calcium chloride or admixtures containing calcium chloride shall not be used in the concrete.

941.02.02 Concrete Mix Design – Concrete shall meet the requirements of Table 941-A.

Table 941-A: Concrete Mix Requirements

Property	Criteria
Minimum compressive strength at 28 days	32 MPa
Nominal Maximum Size of Coarse Aggregate (NMSA)	20 mm or 28 mm
Air Content – 20mm NMSA	5% to 8%
Air Content – 28mm NMSA	4% to 7%
Slump	120 ± 30 mm
Class of Exposure	C-2

941.02.03 Self-Consolidating Concrete – Self-Consolidating Concrete (SCC) may be used in the manufacture of precast concrete barrier, but only under the following conditions:

(a) The self-consolidating concrete mix shall conform to the requirements of CSA A23.1, including Clause 8.6 and Table 22, and

(b) The Contractor shall submit the following for review upon request by the Ministry Representative:

(i) a full copy of the applicable concrete mix design utilized in the manufacture of precast concrete barrier, which may include a slump greater than that specified in Table 941-A; and

(ii) Documentation in accordance with either of the following:

(A) Details in accordance with CSA A23.2-24C documenting the proven history of successful performance for the required strength, durability, and other performance requirements of the proposed mix design in the manufacture of precast concrete barrier, or

(B) A full-scale test shall be used to verify the self-consolidating characteristics for placement and for the hardened concrete properties of the mix design for the precast

concrete barrier. Documentation in accordance with CSA A23.2-24C shall be submitted demonstrating that the proposed mix design will achieve the required strength, durability, and performance requirements.

941.03 Reinforcing Steel, Fibrillated Fibres, Attachment Hardware & Miscellaneous Items

941.03.01 Steel Mesh Reinforcement – Welded steel wire mesh reinforcement shall be supplied and installed in each section as shown on the SP941 series Drawings, and in accordance with SS 412. Additional reinforcement may be installed to assist handling during the precasting operations but shall be subject to prior approval by the Ministry Representative.

941.03.02 Fibre Reinforcement – Fibrillated Fibres (polyolefin or polypropylene or a blend of these fibres) are an acceptable substitute of welded wire mesh. Fibrillated fibres shall meet requirements of ASTM C1116 Synthetic Fiber Reinforced Concrete, Type III fibre.

Fibres shall have a minimum length of 50 mm, added at a dosage rate of 1 kg/m^3 (min of 0.1% by volume) and shall be thoroughly mixed with concrete before placement into the forms.

Fibres shall have a minimum tensile strength of 350 MPa and a minimum modulus of elasticity of 4.2 GPa.

Fibres are to be added early in the mixing process following the manufacturer's recommendations to ensure evenly distributed fibres.

The supplier/manufacturer of the fibre must furnish test data in accordance with ASTM C1399 to the precast concrete manufacturer to show the fibre complies with the specification requirements as part of the precast concrete manufacturer's quality control plan.

When the fibre option is used, a single length of 15 mm rebar shall be wire tied to the horizontal sections of the hook or eye assemblies as shown on the SP941 series Drawings.

Steel fibres shall not be used in the concrete mix for concrete barriers.

941.03.03 Reinforcing Steel Bars – Reinforcing steel for bent and hooked connections shall conform to CSA G40.21-M Grade 260W and shall be carefully bent to the radii detailed and installed as shown on the SS Drawings.

The steel manufacturer shall provide, to the precast concrete manufacturer, a certificate giving the results of each of the mechanical and chemical tests applicable to the steel material produced, confirming that the material has been produced in accordance with the specified requirements. This certificate shall be provided to the Ministry Representative upon request.

Bending shall be done by methods that will not produce fracture or other injury. The metal heating shall not be to a higher temperature than that producing a "dark cherry red" colour. After heating, the metal shall be cooled as slowly as possible. Following the bending, the surface of the metal shall be carefully inspected for evidence of fracture, and any fractured pieces shall be replaced.

Prior to delivery, exposed surfaces of connections shall be prepared for and given a heavy application of zinc rich coating selected from the Ministry's Recognized Products List under the category of "Additional Paint Coatings – Zinc-Rich Touch-up Paint and Primers".

941.03.04 Pick-up Point Forming – Pick-up points for handling units shall be formed with accurately placed rigid P.V.C. pipe recessed 15 mm from both finished surfaces as detailed.

941.03.05 Optional Features – Where barrier is ordered with drainage slots or grouting holes or both, they shall be accurately cast-in as detailed. Facilities for the installation of anti-glare screens will be detailed as and when required.

941.04 Placing and Finishing of Concrete

941.04.01 Placing and Consolidating – Concrete shall be placed in the forms and carefully consolidated in strict accordance with CSA A23.4, Clause 21.

941.04.02 Curing and Protection – Curing and protection shall be carried out strictly according to CSA A23.4, Clause 23 and as follows:

(a) Curing shall be considered complete when test cylinders reach the specified 28-day compressive strength provided such strength is reached not later than 28 days after the barriers are cast.

(b) Steam curing is permissible for either the entire curing period or portion thereof and shall be carried out in accordance with CSA A23.4, Clause 23.2.3.

(c) At no time during or at completion of the curing period shall the temperature differential between the concrete surface and the ambient temperature be greater than 20°C.

(d) If steam is used for a portion of the curing period, additional type 2 curing shall be carried out after the steam curing, according to CSA A23.4 Clause 23 until such time as strength tests of concrete test cylinders that have been both steam and normally cured with the barriers reach the specified concrete strength.

941.04.03 Exposed Surfaces – Exposed surfaces shall be uniform in texture and colour as produced from well-maintained steel form surfaces and proper vibration methods without excessive surface fines or laitance.

941.04.04 Defects – Defects of the exposed surface will normally be cause for the rejection of any unit except where such are within the permissible limits or are subject to making good as follows:

(a) Unobtrusive defects of any kind where their total area is not in excess of 2% of the exposed surface area of the unit.

(b) Air holes not greater than 3 mm in diameter and not more than twenty (20) holes in any isolated 300 mm X 300 mm area.

(c) Sharp protrusions at the edges of the exposed surfaces where necessary shall be softened by careful rubbing or grinding.

(d) Patching of isolated small holes, cavities and similar self-confining defects may be permitted when authorized in writing by the Ministry Representative.

941.04.05 Patching – Patching, only when authorized by the Ministry Representative, shall have the defective area well saturated with water and the defect prepared with cement paste and filled with mortar. The mortar, properly proportioned of the same sand and cement as the original concrete and reasonably colour matched to the cured dry unit with the addition of white cement where necessary, shall be pre-shrunk for about one hour before retempering and use. The patching mortar shall be well tooled in, finished flush and smooth and the area covered to cure adequately.

941.04.06 All concrete surfaces prior to shipment shall be accurate to detail and, in particular at the end connections, true to dimension tolerances. End connection surfaces shall be cleared out of any out-of-tolerance material, to allow a secure connection between components..

941.05 Tolerances – Allowable Tolerances for the concrete dimensions of the barriers shall be ± 3 mm except as otherwise indicated on the detail drawings.

941.06 Procedure of Manufacture – The Contractor (or the manufacturer, if the Ministry is purchasing directly) shall notify the Ministry in advance concerning the date when the order is to be manufactured, so that the Ministry Quality Assurance may be carried out. All processes shall be open for inspection and approval by the Ministry Representative.

Material shipped prior to the inspection or written release for shipment by the Ministry Representative shall be at the Contractor's risk (or supplier's risk, where the Ministry is purchasing directly).

The manufacturer's name or trademark, year of manufacture and form number shall be embedded on the end of each unit in a manner, size and depth that are permanently legible.

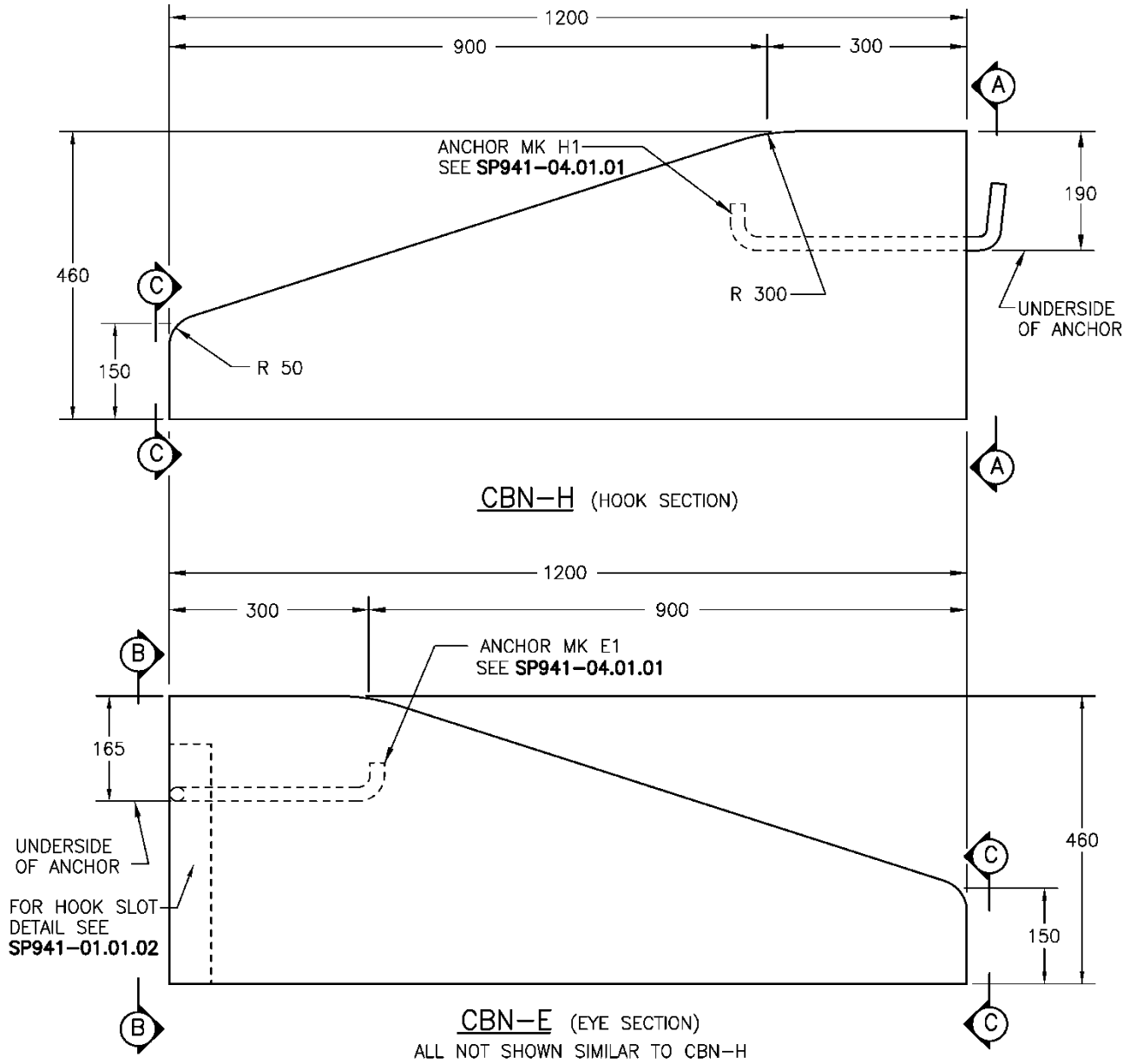
Authorized patching or making good may be inspected before shipment or upon delivery and the rejected unit(s) shall be replaced at no cost to the Ministry.

941.07 Handling – In handling the finished product, the concrete and connecting devices shall not be damaged or distorted.

941.08 Payment – Payment shall be at Unit Price bid per unit (section). The price bid shall be full compensation for everything furnished and done including, but not limited to, supply of forms and all materials, placing, vibrating, curing, quality control, concrete quality control testing, stripping, finishing, general clean-up and delivery.

PRECAST CONCRETE BULL–NOSE 460 mm – CBN–H & CBN–E

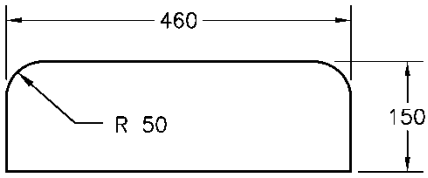
SP941–01.01.01



ELEVATION

GENERAL NOTES:

1. FOR VIEW (A) & (B) SEE **SP941–01.01.02**
2. FIBRILATED FIBRE STRAND REINFORCED CONCRETE TO BE USED FOR BULL–NOSE (CBN).
3. FIBRILLATED FIBRE STRAND REINFORCED CONCRETE MAY BE SUBSTITUTED FOR STEEL REINFORCED CONCRETE IN LOW BARRIER (CLB–H & E).
4. MATERIALS AND QUALITY OF WORK TO BE IN ACCORDANCE WITH SECTION **941.**
5. STANDARD CBN–E MAY BE MANUFACTURED ONLY WITH PRIOR WRITTEN PERMISSION FROM THE HIGHWAY SAFETY BRANCH.

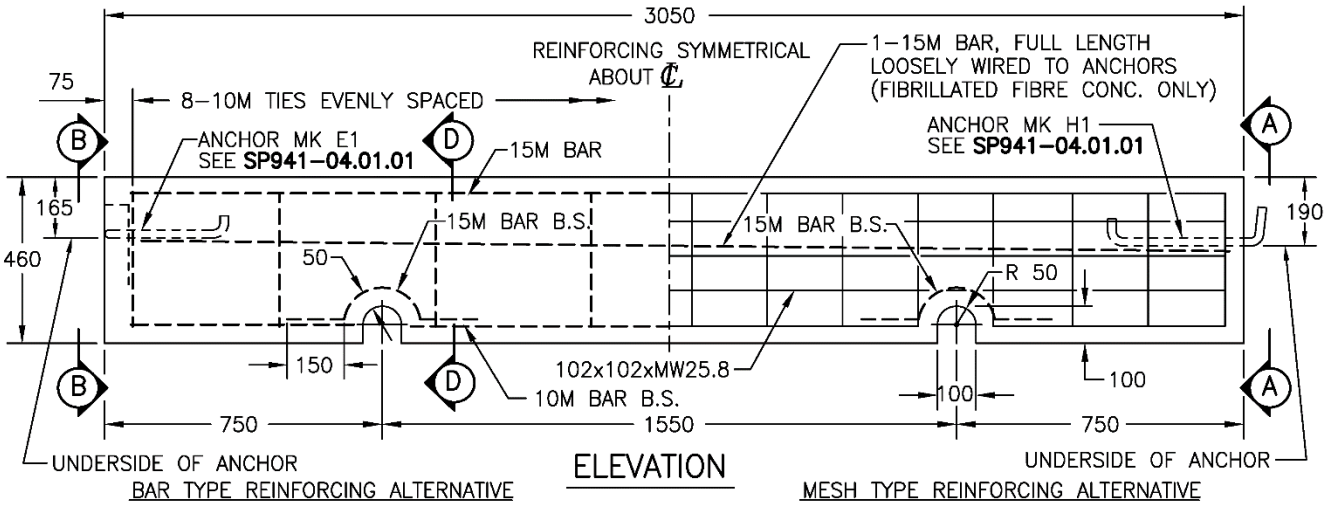


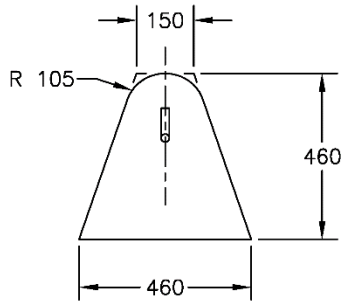
VIEW (C)

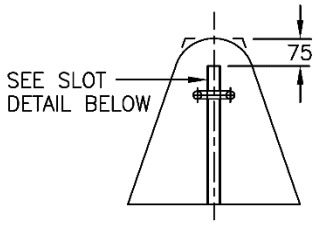
NOT TO SCALE ALL DIMENSIONS ARE IN MILLIMETRES UNLESS OTHERWISE NOTED

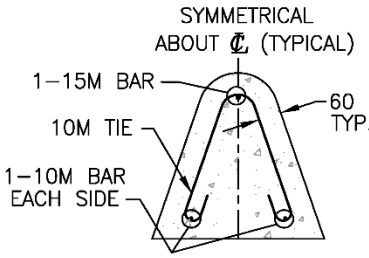
PRECAST CONCRETE LOW BARRIER 460 mm – CLB–E+H

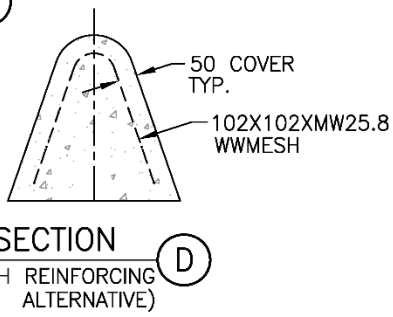
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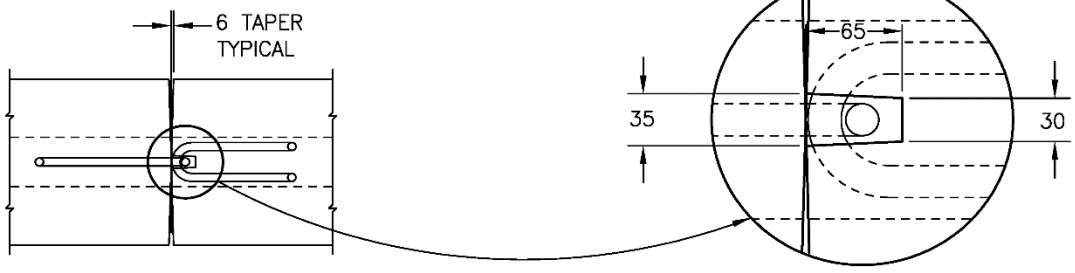








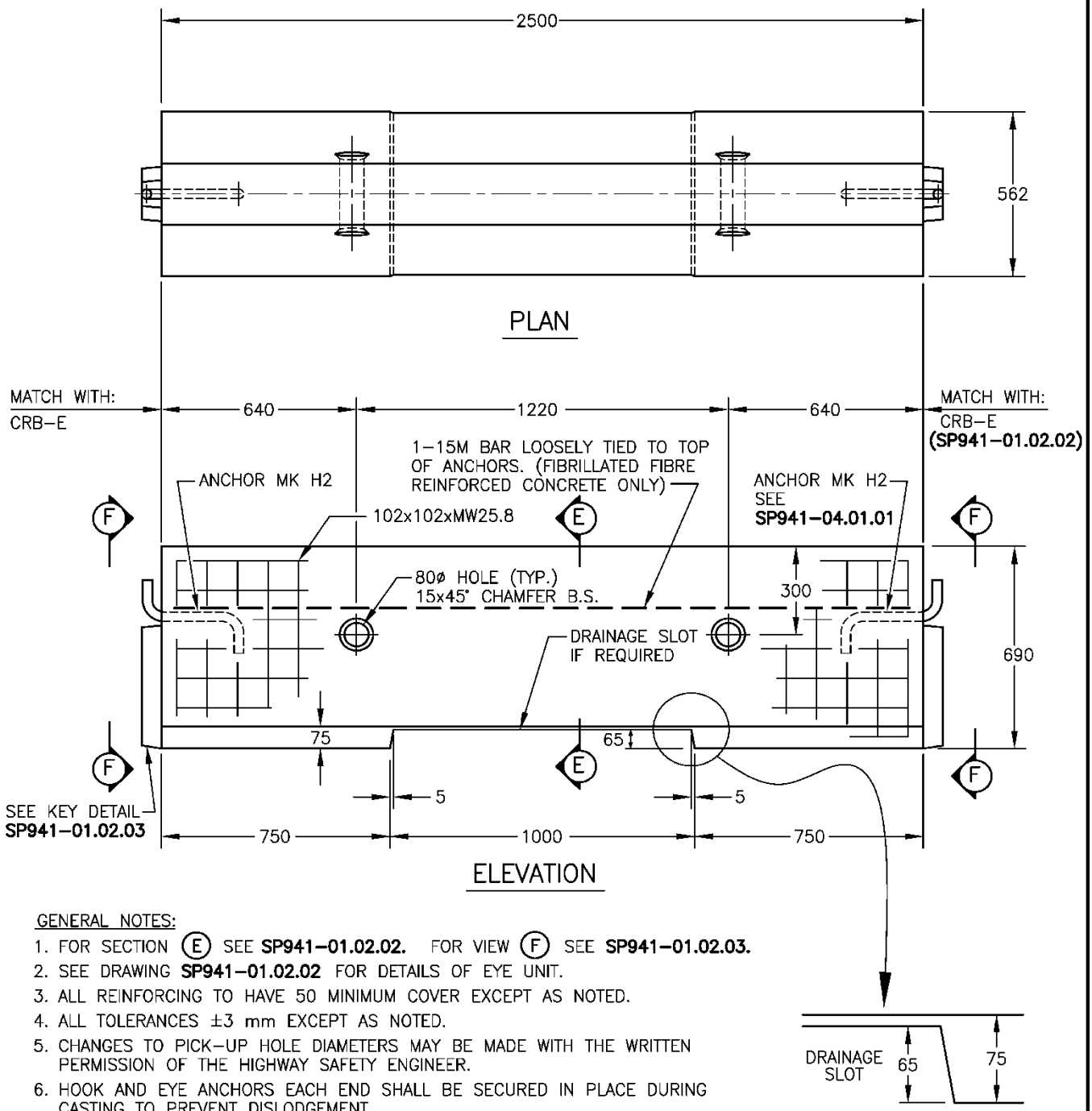




NOTE: FOR GENERAL NOTES SEE **SP941–01.01.01**

NOT TO SCALE ALL DIMENSIONS ARE IN MILLIMETRES UNLESS OTHERWISE NOTED

PRECAST CONCRETE ROADSIDE BARRIER 690 mm – CRB–H	SP941–01.02.01



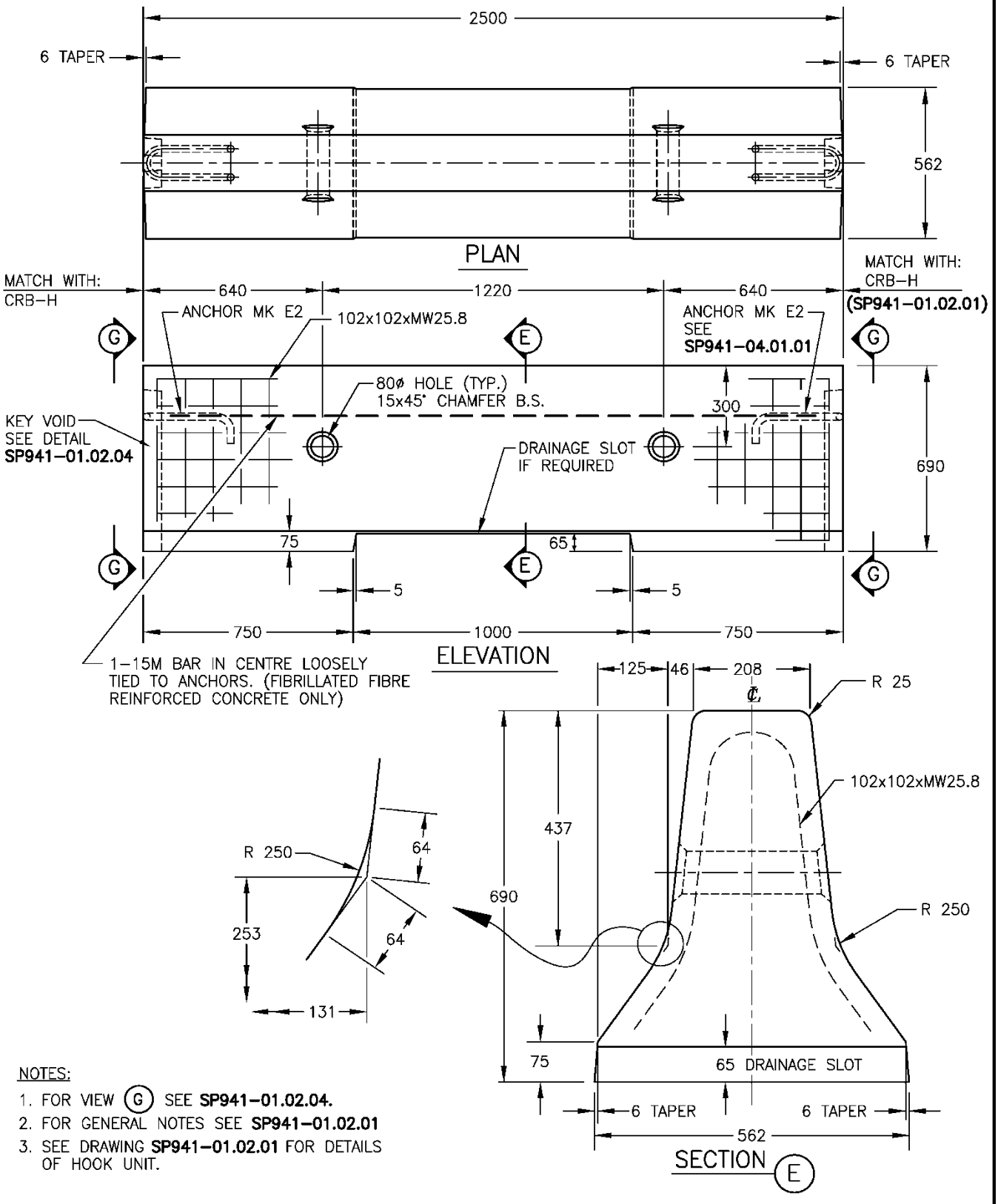
GENERAL NOTES:

1. FOR SECTION (E) SEE **SP941–01.02.02.** FOR VIEW (F) SEE **SP941–01.02.03.**
2. SEE DRAWING **SP941–01.02.02** FOR DETAILS OF EYE UNIT.
3. ALL REINFORCING TO HAVE 50 MINIMUM COVER EXCEPT AS NOTED.
4. ALL TOLERANCES ±3 mm EXCEPT AS NOTED.
5. CHANGES TO PICK–UP HOLE DIAMETERS MAY BE MADE WITH THE WRITTEN PERMISSION OF THE HIGHWAY SAFETY ENGINEER.
6. HOOK AND EYE ANCHORS EACH END SHALL BE SECURED IN PLACE DURING CASTING TO PREVENT DISLODGEMENT.
7. MATERIALS AND QUALITY OF WORK TO BE IN ACCORDANCE WITH SECTION **941.**
8. DRAINAGE SLOT IS REQUIRED WHEN NECESSARY TO DRAIN SURFACE WATER THROUGH THE BARRIER.
9. SHEAR KEY VOID WITH GROUTING HOLES BY REQUEST. SEE **SP941–04.02.01.**
10. FIBRILLATED FIBRE STRAND REINFORCED CONCRETE MAY BE SUBSTITUTED FOR WELDED WIRE MESH REINFORCED CONCRETE.

NOT TO SCALE ALL DIMENSIONS ARE IN MILLIMETRES UNLESS OTHERWISE NOTED

PRECAST CONCRETE ROADSIDE BARRIER 690 mm – CRB–E

SP941–01.02.02



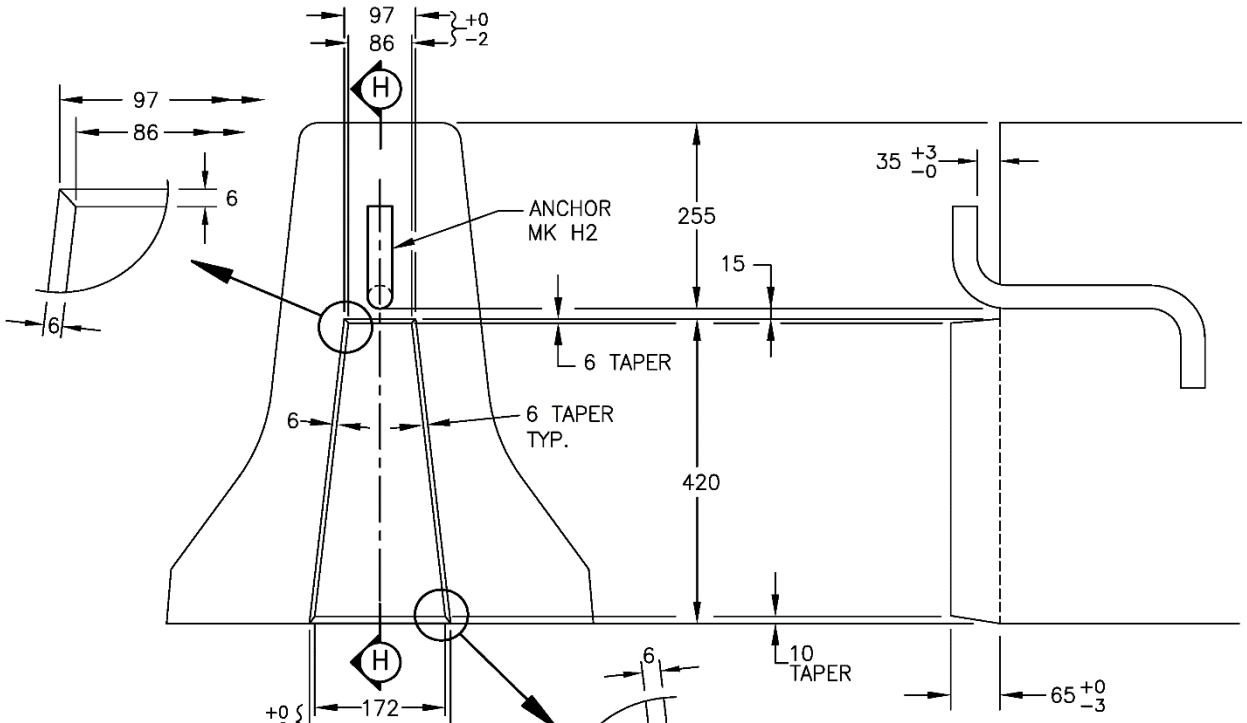
NOTES:

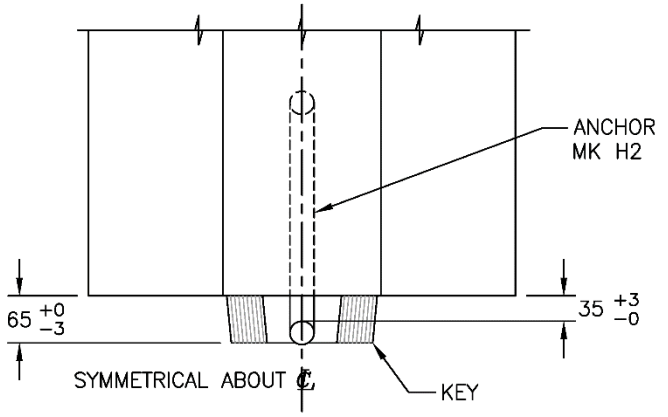
1. FOR VIEW (G) SEE **SP941–01.02.04.**
2. FOR GENERAL NOTES SEE **SP941–01.02.01**
3. SEE DRAWING **SP941–01.02.01** FOR DETAILS OF HOOK UNIT.

NOT TO SCALE ALL DIMENSIONS ARE IN MILLIMETRES UNLESS OTHERWISE NOTED

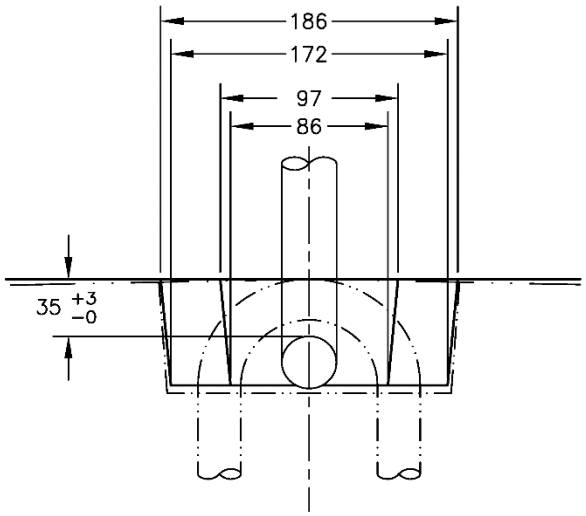
PRECAST CONCRETE ROADSIDE BARRIER 690 mm – CRB–H DETAILS

SP941–01.02.03





PLAN – END – HOOK UNIT



CONNECTION DETAIL

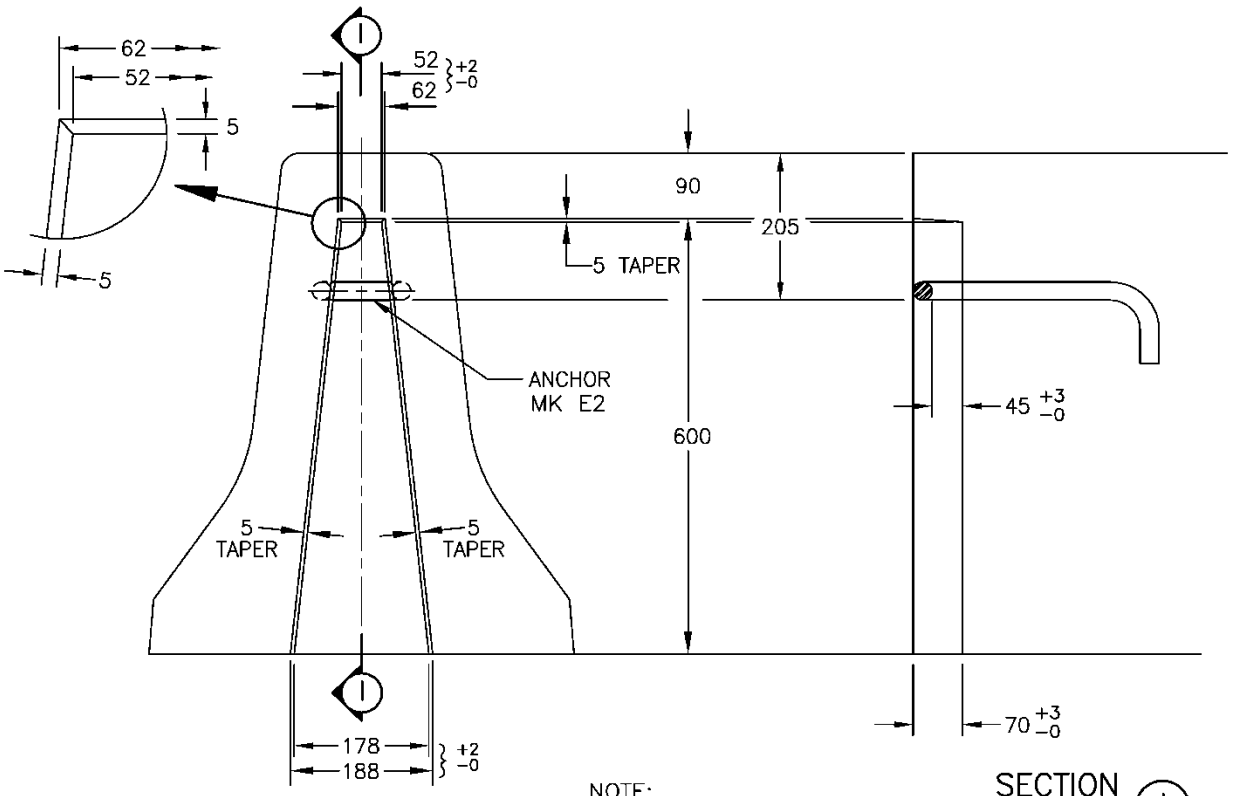
NOTES:

1. FOR LOCATION OF VIEW (F) SEE **SP941–01.02.01, SP941–03.02.01, SP941–03.03.01 & SP941–03.03.03.**
2. FOR GENERAL NOTES SEE **SP941–01.02.01**
3. FOR DETAILS OF ANCHORS SEE **SP941–04.01.01.**
4. THE CONNECTION DIMENSIONS ON A CRB–E UNIT PERMIT INSTALLATION TO A MINIMUM HORIZONTAL CURVE RADIUS OF 51 METRES. FOR TIGHT CURVATURE INSTALLATION SEE **SP941–02.01.08 & SP941–02.01.09.**

NOT TO SCALE ALL DIMENSIONS ARE IN MILLIMETRES UNLESS OTHERWISE NOTED

PRECAST CONCRETE ROADSIDE BARRIER 690 mm – CRB–E DETAILS

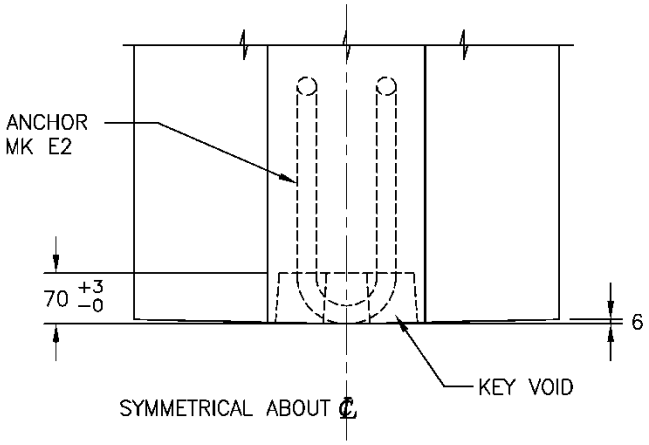
SP941–01.02.04



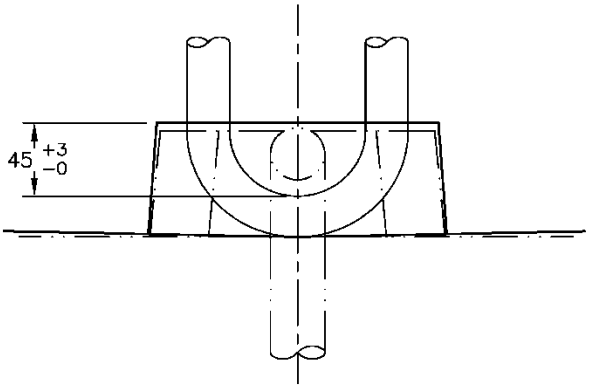
VIEW (EYE UNIT) G

NOTE:
FOR DIMENSIONS NOT SHOWN SEE SECTION E **SP941–01.02.02**

SECTION I



PLAN – END – EYE UNIT



CONNECTION DETAIL

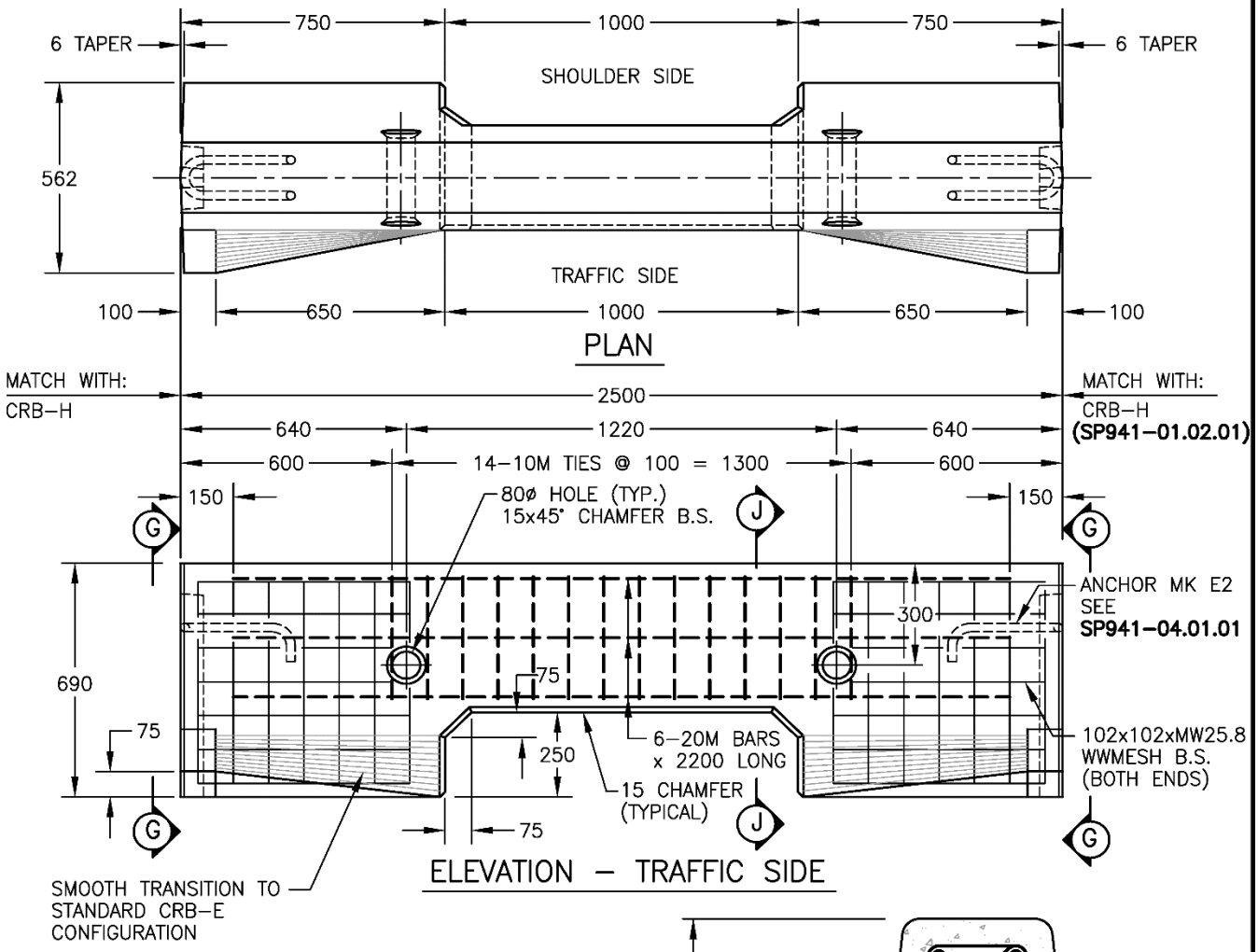
NOTES:

1. FOR LOCATION OF VIEW (G) SEE **SP941–01.02.02, SP941–01.02.05 & SP941–03.01.01.**
2. FOR GENERAL NOTES SEE **SP941–01.02.01.**
3. FOR DETAILS OF ANCHORS SEE **SP941–04.01.01.**
4. THE CONNECTION DIMENSIONS ON A CRB–E UNIT PERMIT INSTALLATION TO A MINIMUM HORIZONTAL CURVE RADIUS OF 51 METRES. FOR TIGHT CURVATURE INSTALLATION SEE **SP941–02.01.08 & SP941–02.01.09.**

NOT TO SCALE ALL DIMENSIONS ARE IN MILLIMETRES UNLESS OTHERWISE NOTED

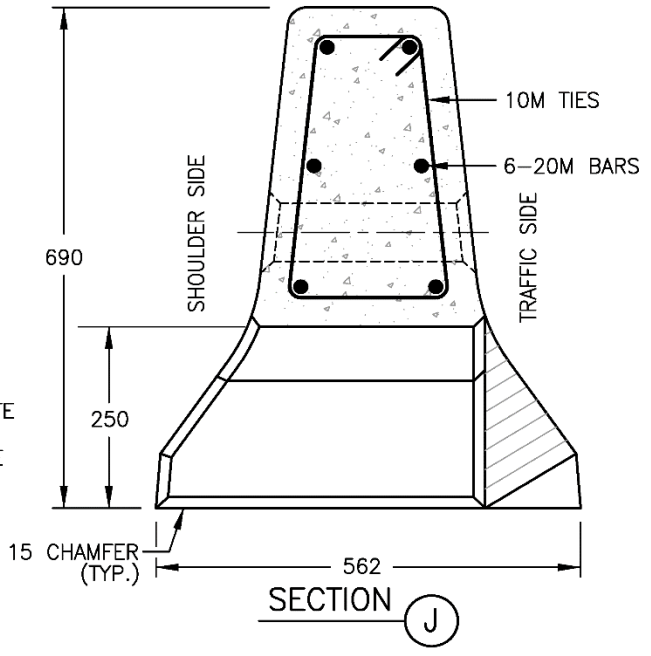
PRECAST CONCRETE DRAINAGE BARRIER (SPECIAL USE[5]) 690 mm – CDB–E DETAILS

SP941–01.02.05



NOTES:

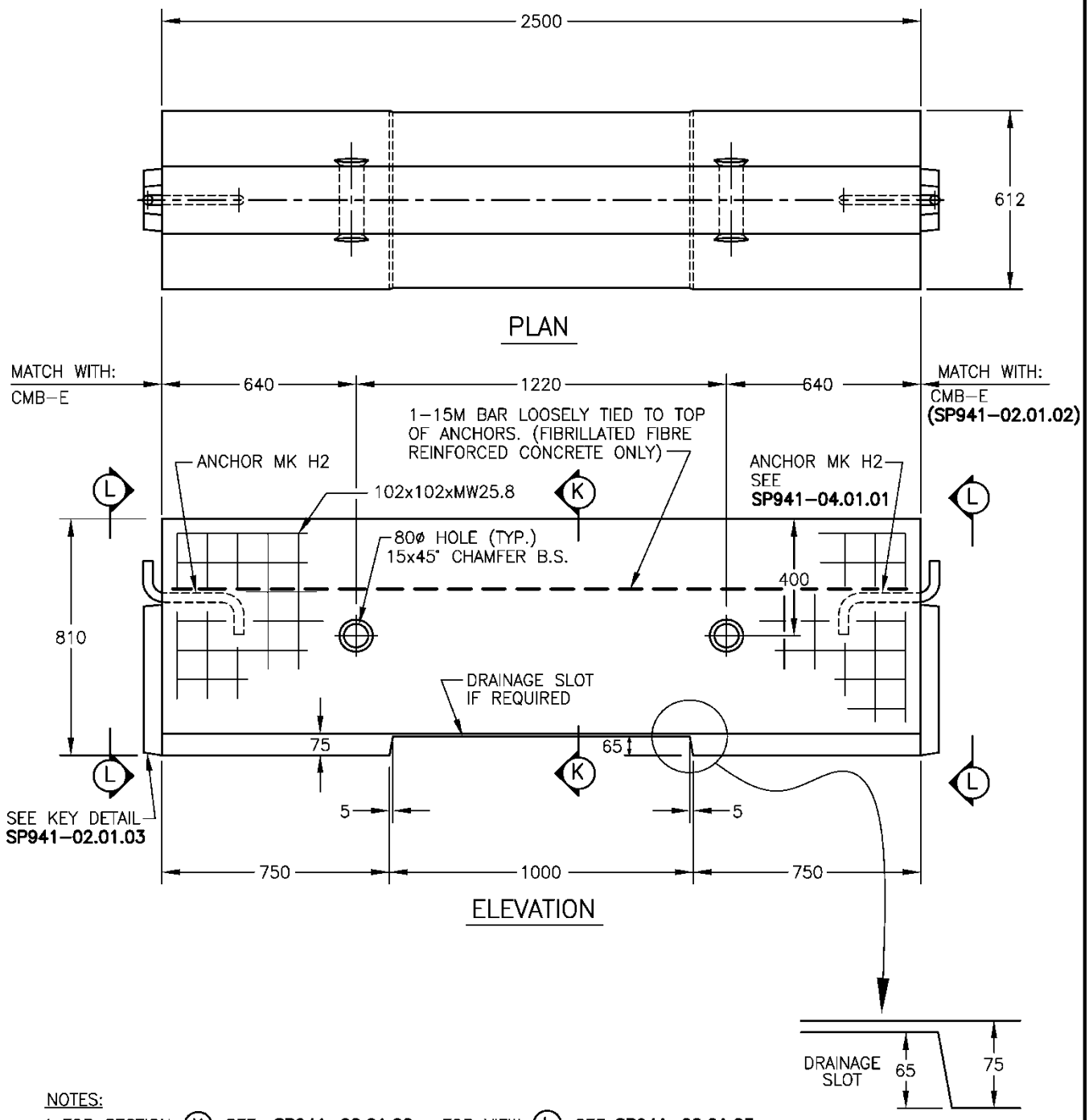
1. FOR VIEW (G) SEE **SP941–01.02.04.**
2. SEE **SP941–01.02.01** FOR DETAILS OF CONNECTING HOOK UNIT.
3. FOR GENERAL NOTES SEE **SP941–01.02.01**
4. FIBRILLATED FIBRE STRAND REINFORCED CONCRETE MAY BE SUBSTITUTED FOR WELDED WIRE MESH REINFORCED CONCRETE. ADDITIONAL BARS AND TIES WILL STILL APPLY.
5. THIS LARGE OPENING DRAINAGE BARRIER WAS DEVELOPED TO BE USED IN CONJUNCTION WITH A PAVED SPILLWAY AT THE BOTTOM OF SAG CURVES. IT MAY BE APPROPRIATE FOR USE AT OTHER LOCATIONS WHERE A SIGNIFICANT VOLUME OF WATER REQUIRES AN OUTLET THAT CANNOT BE HANDLED BY CRB WITH A 65 mm DRAINAGE SLOT.



NOT TO SCALE ALL DIMENSIONS ARE IN MILLIMETRES UNLESS OTHERWISE NOTED

PRECAST CONCRETE MEDIAN BARRIER 810 mm – CMB–H

SP941–02.01.01



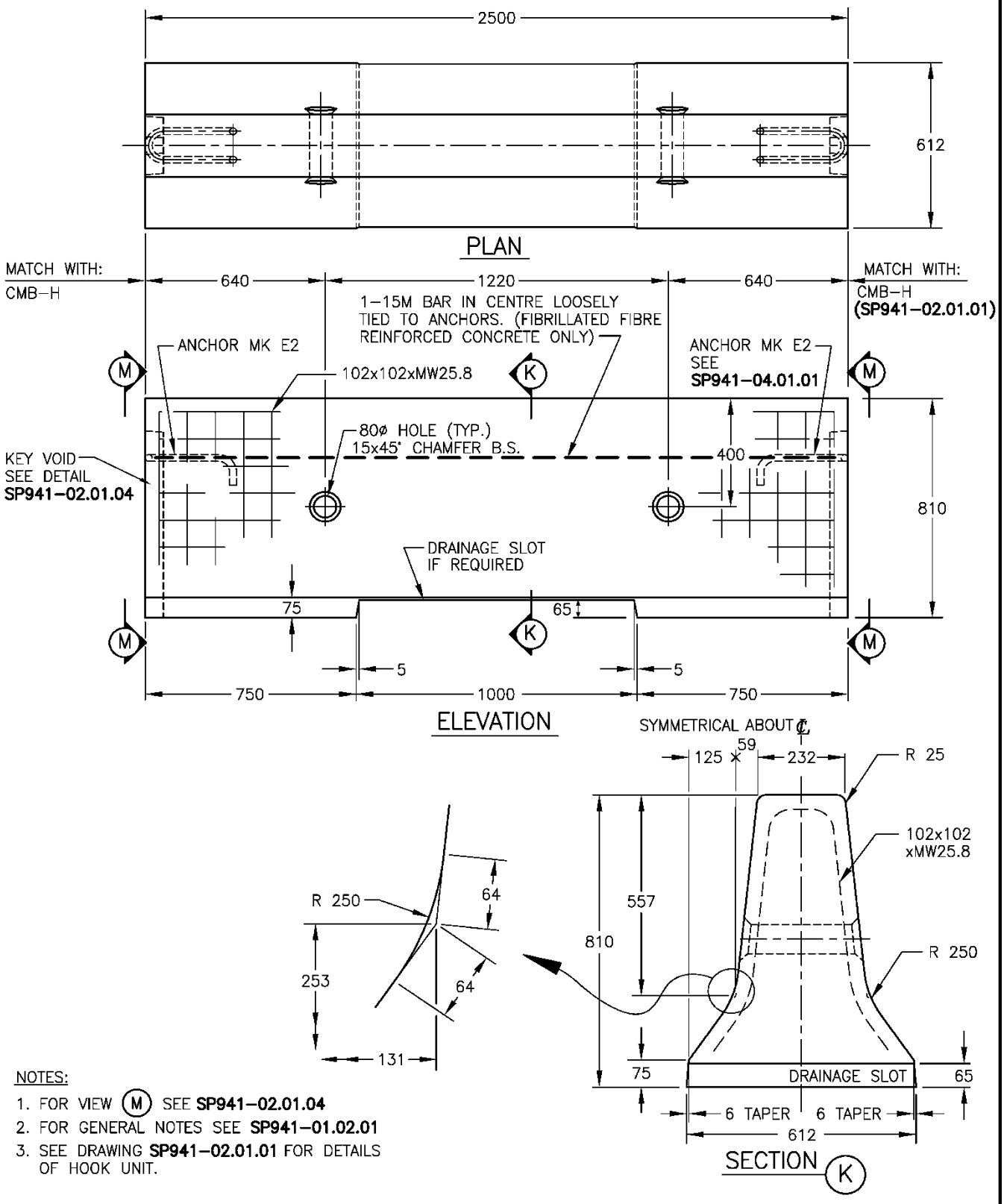
NOTES:

1. FOR SECTION (K) SEE **SP941–02.01.02** FOR VIEW (L) SEE **SP941–02.01.03.**
2. FOR GENERAL NOTES SEE **SP941–01.02.01**
3. SEE DRAWING **SP941–02.01.02** FOR DETAILS OF EYE UNIT.

NOT TO SCALE ALL DIMENSIONS ARE IN MILLIMETRES UNLESS OTHERWISE NOTED

PRECAST CONCRETE MEDIAN BARRIER 810 mm – CMB–E

SP941–02.01.02



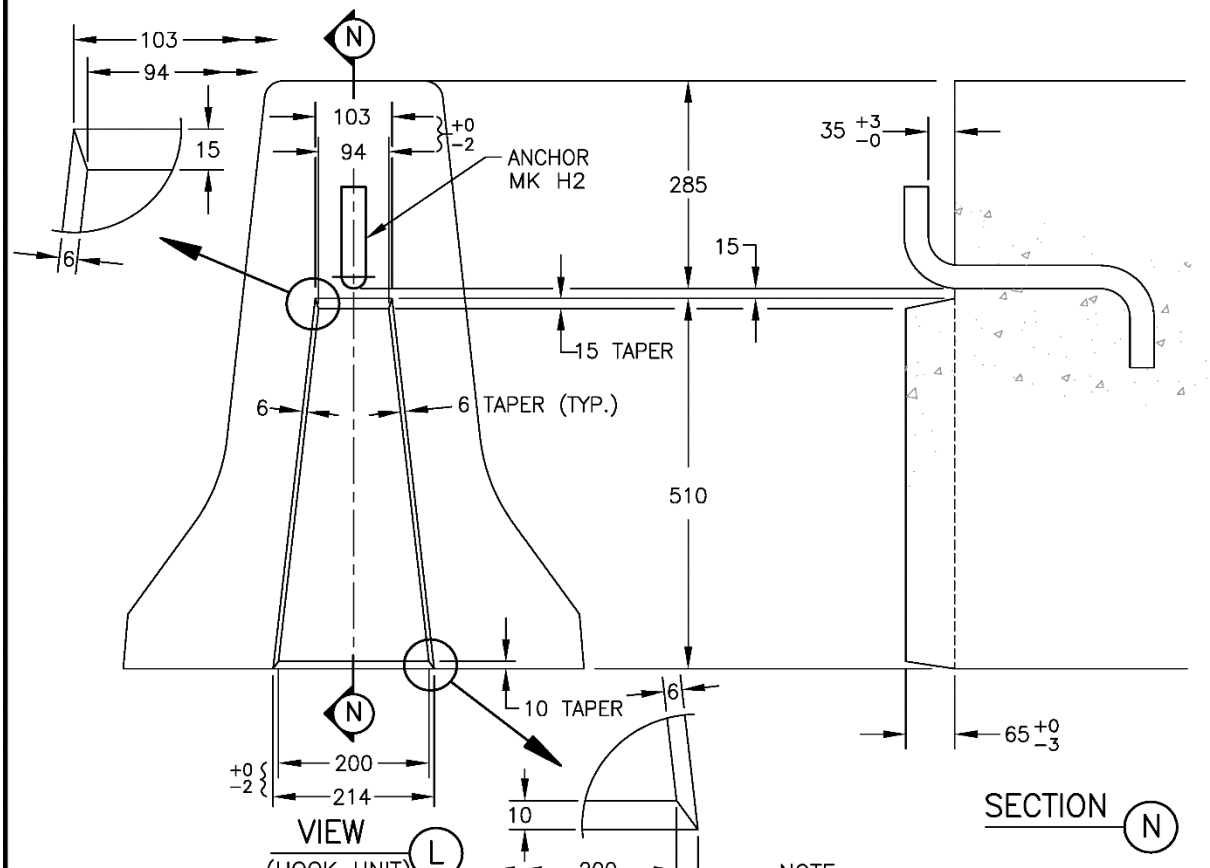
NOTES:

1. FOR VIEW (M) SEE **SP941–02.01.04**
2. FOR GENERAL NOTES SEE **SP941–01.02.01**
3. SEE DRAWING **SP941–02.01.01** FOR DETAILS OF HOOK UNIT.

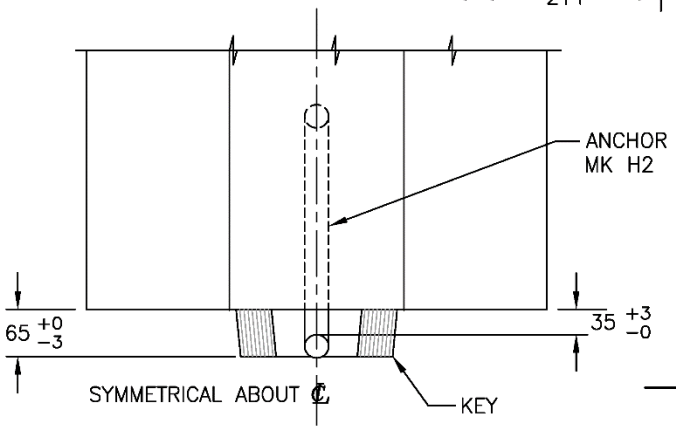
NOT TO SCALE ALL DIMENSIONS ARE IN MILLIMETRES UNLESS OTHERWISE NOTED

PRECAST CONCRETE MEDIAN BARRIER 810 mm – CMB–H DETAILS

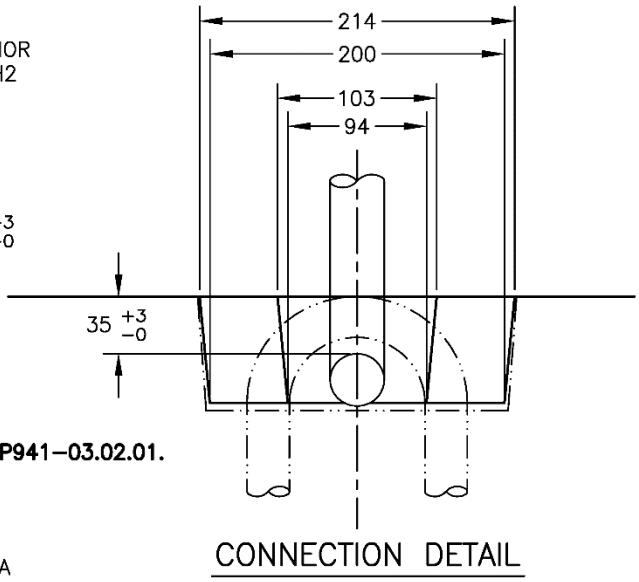
SP941–02.01.03



NOTE:
FOR DIMENSIONS NOT SHOWN SEE SECTION (K) **SP941–02.01.02**



PLAN – END – HOOK UNIT



CONNECTION DETAIL

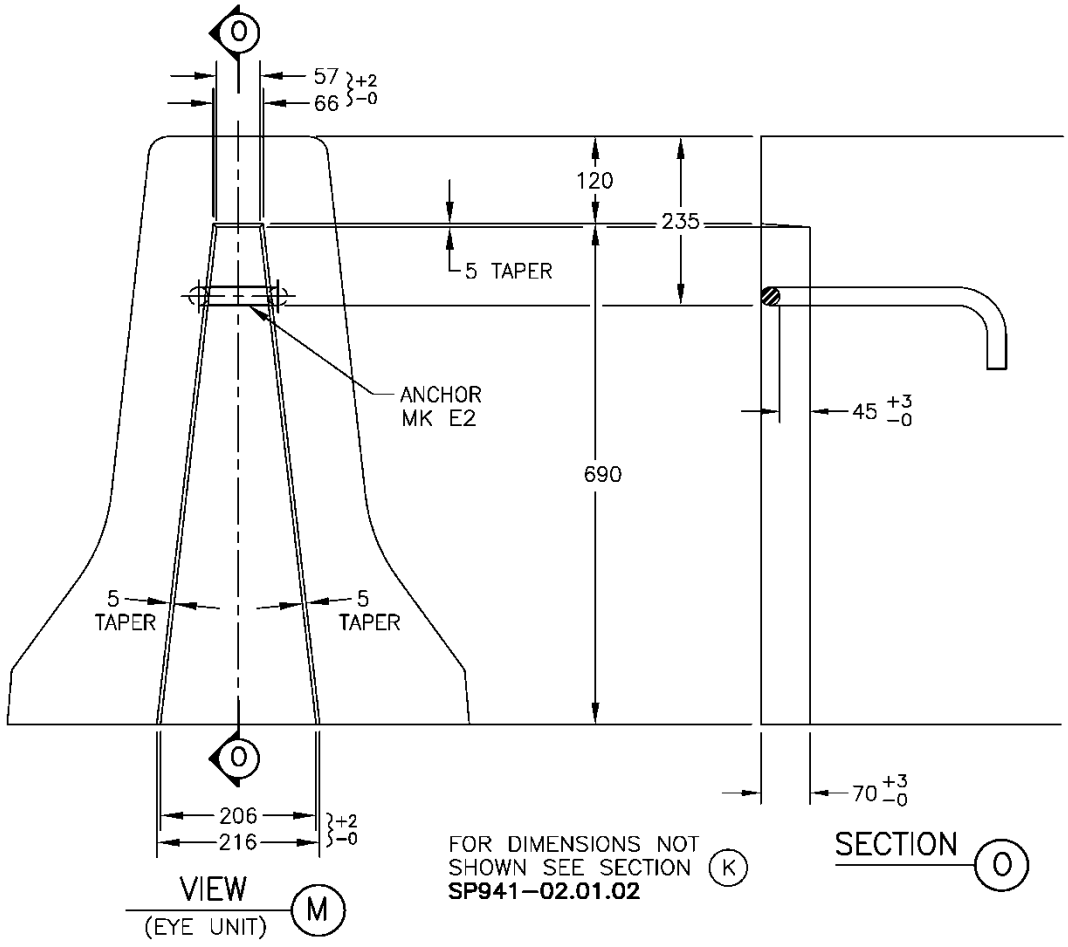
NOTES:

1. FOR LOCATION OF VIEW (L) SEE **SP941–02.01.01 & SP941–03.02.01.**
2. FOR GENERAL NOTES SEE **SP941–01.02.01.**
3. FOR DETAILS OF ANCHORS SEE **SP941–04.01.01.**
4. THE CONNECTION DIMENSIONS PERMIT INSTALLATION TO A MINIMUM HORIZONTAL CURVE RADIUS OF 95.5 METRES.

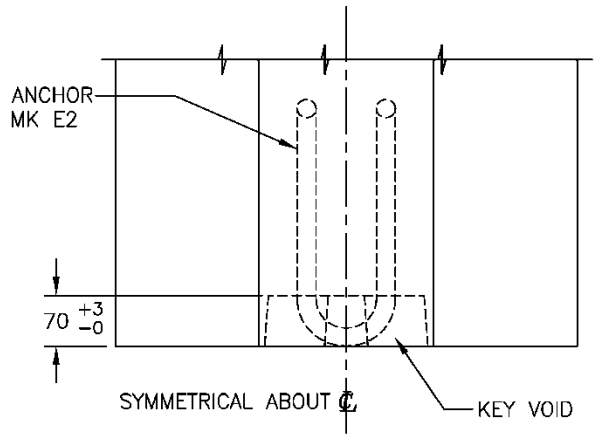
NOT TO SCALE ALL DIMENSIONS ARE IN MILLIMETRES UNLESS OTHERWISE NOTED

PRECAST CONCRETE MEDIAN BARRIER 810 mm – CMB–E DETAILS

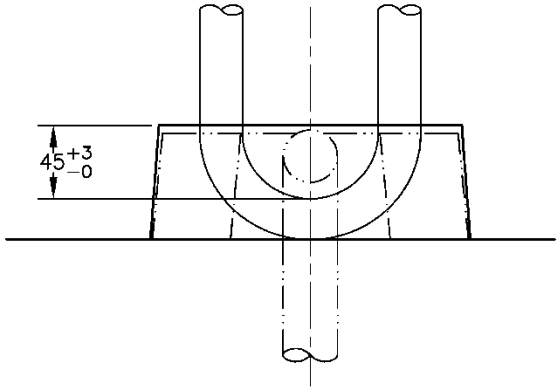
SP941–02.01.04



VIEW (EYE UNIT) M



PLAN – END – EYE UNIT



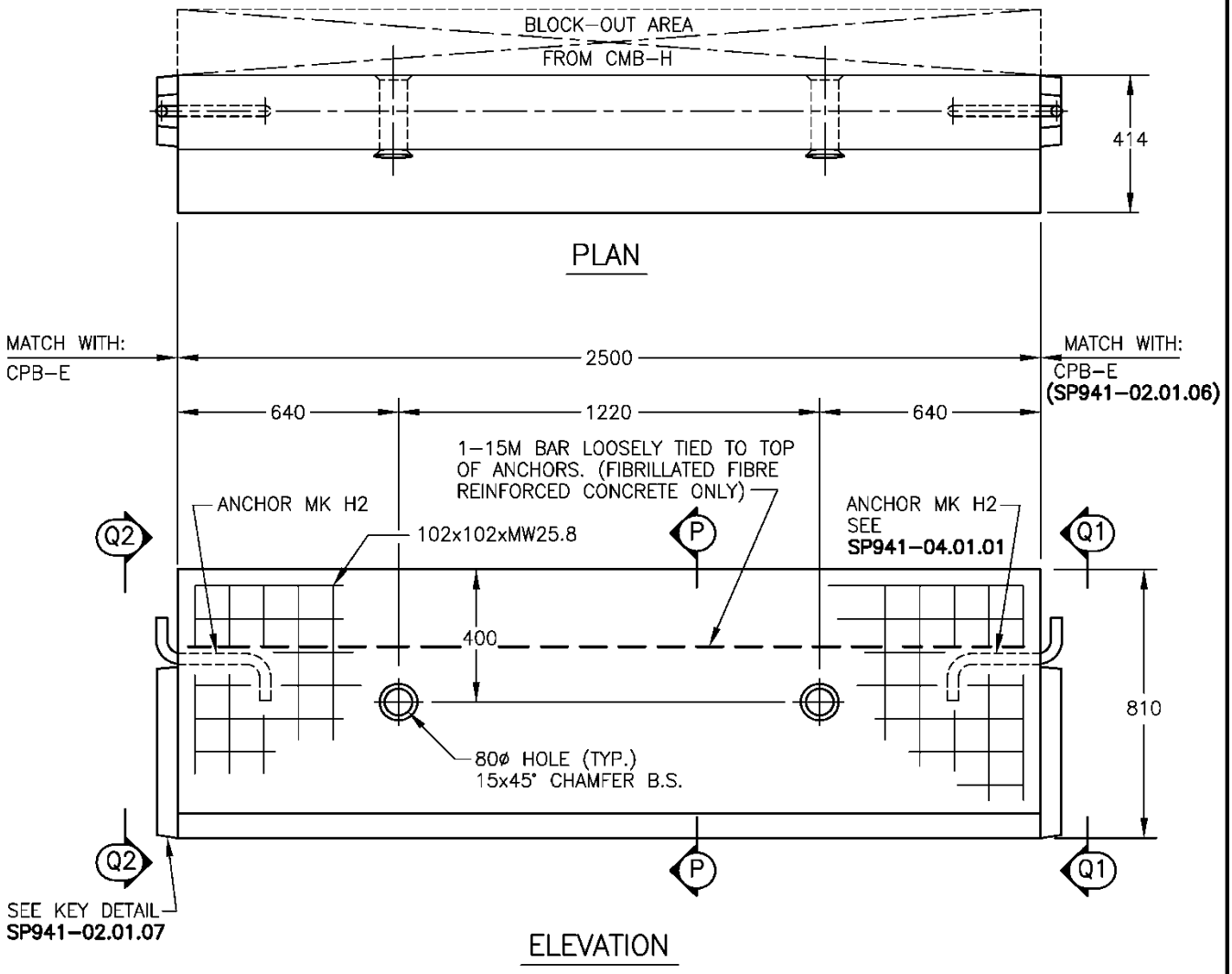
CONNECTION DETAIL

NOTES:

1. FOR LOCATION OF VIEW (M) SEE **SP941–02.01.02.**
2. FOR GENERAL NOTES SEE **SP941–01.02.01.**
3. FOR DETAILS OF ANCHORS SEE **SP941–04.01.01.**
4. THE CONNECTION DIMENSIONS PERMIT INSTALLATION TO A MINIMUM HORIZONTAL CURVE RADIUS OF 95.5 METRES.

NOT TO SCALE ALL DIMENSIONS ARE IN MILLIMETRES UNLESS OTHERWISE NOTED

PRECAST CONCRETE PIER BARRIER 810 mm – CPB–H	SP941–02.01.05



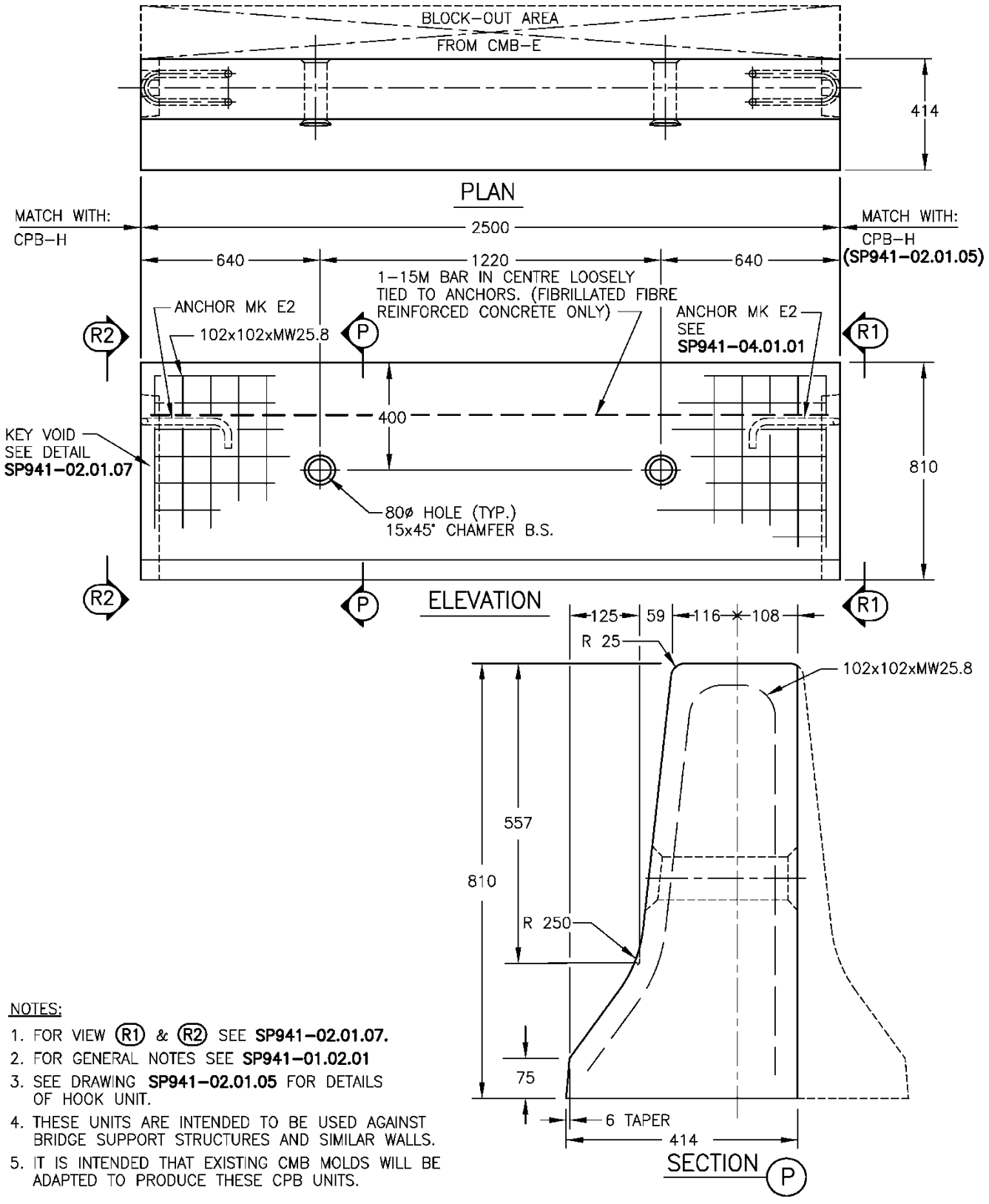
NOTES:

1. FOR SECTION (P) SEE **SP941–02.01.06.**
   FOR VIEW (Q1) & (Q2) SEE **SP941–02.01.07.**
2. FOR GENERAL NOTES SEE **SP941–01.02.01**
3. SEE DRAWING **SP941–02.01.06** FOR DETAILS OF EYE UNIT.
4. THESE UNITS ARE INTENDED TO BE USED AGAINST BRIDGE SUPPORT STRUCTURES AND SIMILAR WALLS.
5. IT IS INTENDED THAT EXISTING CMB MOLDS WILL BE ADAPTED TO PRODUCE THESE CPB UNITS.

NOT TO SCALE ALL DIMENSIONS ARE IN MILLIMETRES UNLESS OTHERWISE NOTED

PRECAST CONCRETE PIER BARRIER 810 mm – CPB–E

SP941–02.01.06



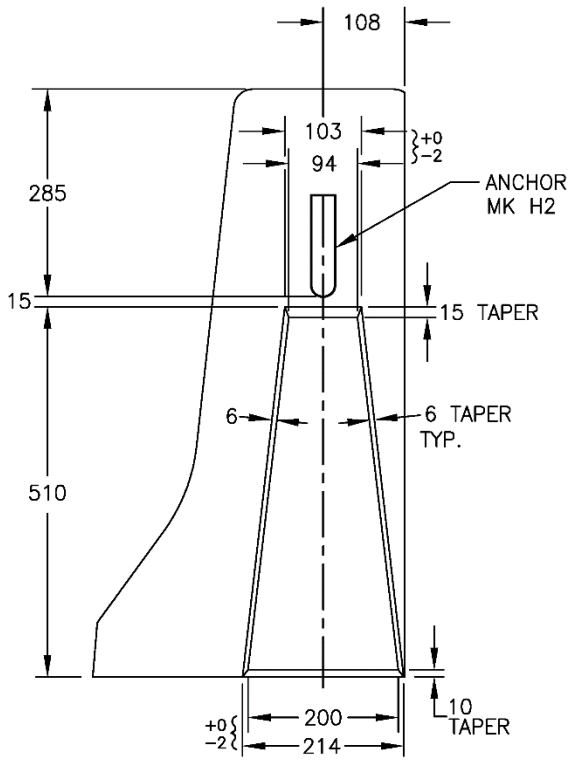
NOTES:

1. FOR VIEW (R1) & (R2) SEE **SP941–02.01.07.**
2. FOR GENERAL NOTES SEE **SP941–01.02.01**
3. SEE DRAWING **SP941–02.01.05** FOR DETAILS OF HOOK UNIT.
4. THESE UNITS ARE INTENDED TO BE USED AGAINST BRIDGE SUPPORT STRUCTURES AND SIMILAR WALLS.
5. IT IS INTENDED THAT EXISTING CMB MOLDS WILL BE ADAPTED TO PRODUCE THESE CPB UNITS.

NOT TO SCALE ALL DIMENSIONS ARE IN MILLIMETRES UNLESS OTHERWISE NOTED

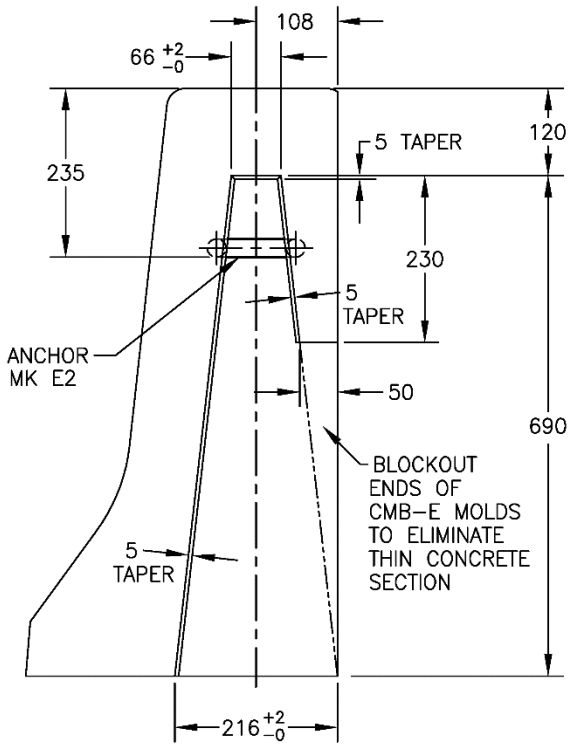
PRECAST CONCRETE PIER BARRIER 810 mm – CPB–H & CPB–E DETAILS

SP941–02.01.07



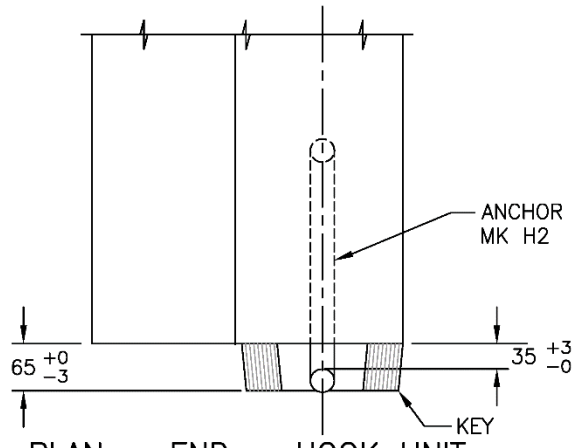
VIEW (HOOK UNIT) Q1 AS SHOWN

VIEW Q2 OPPOSITE HAND

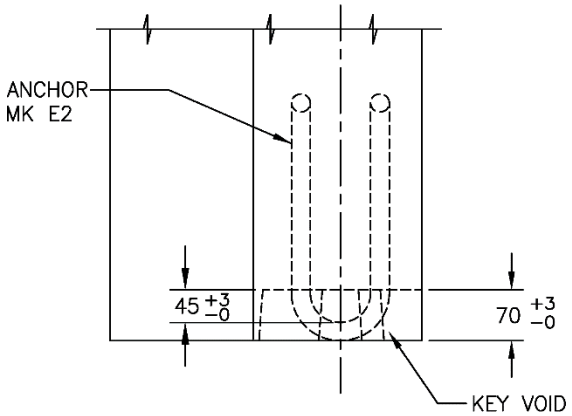


VIEW (EYE UNIT) R1 AS SHOWN

VIEW R2 OPPOSITE HAND



PLAN – END – HOOK UNIT



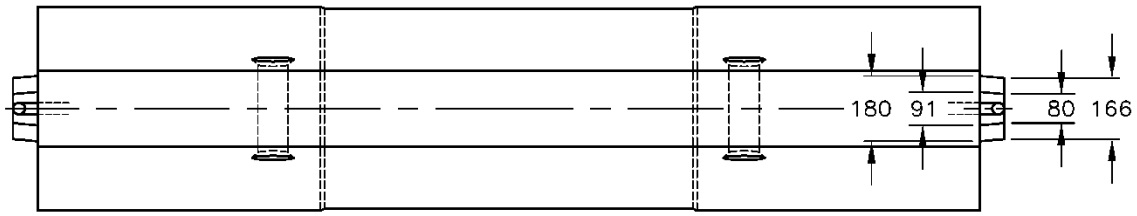
PLAN – END – EYE UNIT

NOTES:

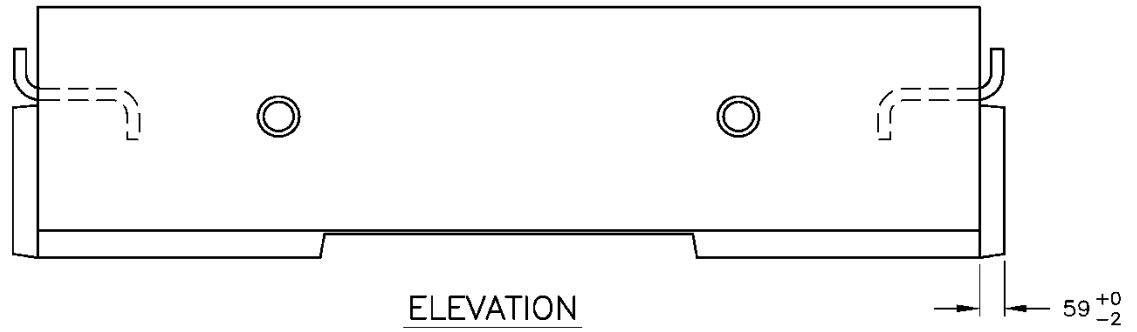
1. FOR LOCATION OF VIEW (Q1) & (Q2) SEE **SP941–02.01.05.**
2. FOR LOCATION OF VIEW (R1) & (R2) SEE **SP941–02.01.06.**
3. FOR GENERAL NOTES SEE **SP941–01.02.01**
4. FOR DETAILS OF ANCHORS SEE **SP941–04.01.01.**

NOT TO SCALE ALL DIMENSIONS ARE IN MILLIMETRES UNLESS OTHERWISE NOTED

PRECAST CRB–H MODIFICATIONS FOR TIGHT CURVATURE INSTALLATION	SP941–02.01.08

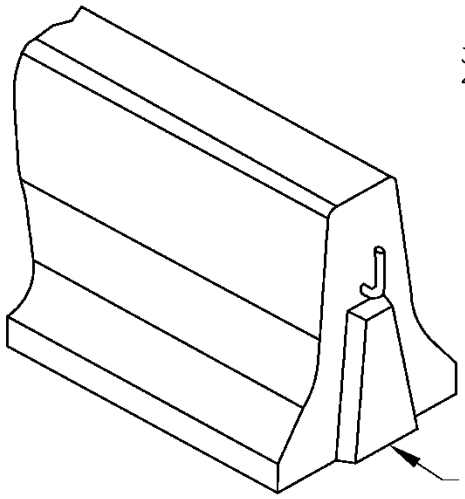


PLAN



ELEVATION

NOTES:
1. THIS DRAWING MUST BE READ WITH CRB DRAWINGS SP941–01.02.01 TO 941–01.02.04 FOR FULL DETAILS OF UNITS.
2. IT IS INTENDED THAT EXISTING CRB MOLDS BE ADAPTED TO DIMENSIONS NOTED ON THIS DRAWING TO PRODUCE UNITS CAPABLE OF BEING PLACED TO A MINIMUM CURVE RADIUS OF APPROXIMATELY 17 m.
3. ALL TOLERANCES AS PER CRB UNIT DRAWINGS.
4. THIS MODIFIED UNIT IS INTENDED TO BE USED ON TIGHT CURVATURE AT INTERSECTIONS ONLY.

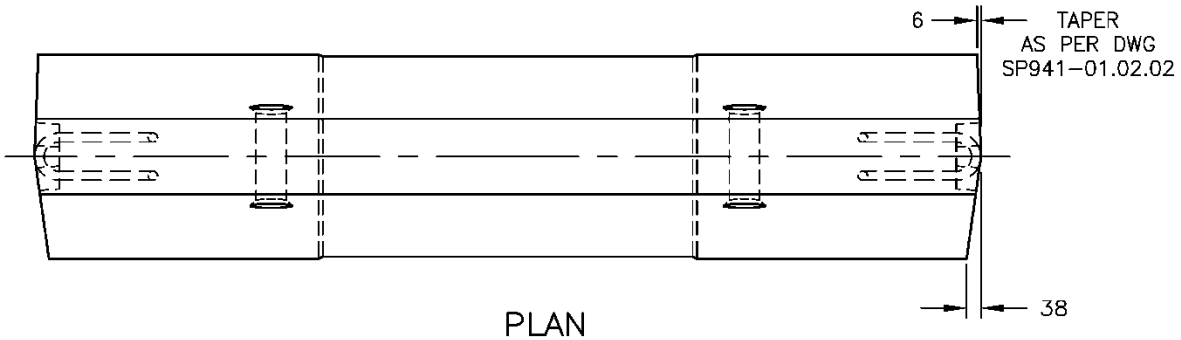


HOOK KEY MADE SMALLER BY USE OF 6 mm PLATE INSERT ON BOTH KEY ENDS, AND 3 mm PLATE INSERTS ON BOTH SIDES OF KEYS DURING CASTING.

NOT TO SCALE ALL DIMENSIONS ARE IN MILLIMETRES UNLESS OTHERWISE NOTED

PRECAST CRB–E MODIFICATIONS FOR TIGHT CURVATURE INSTALLATION

SP941–02.01.09

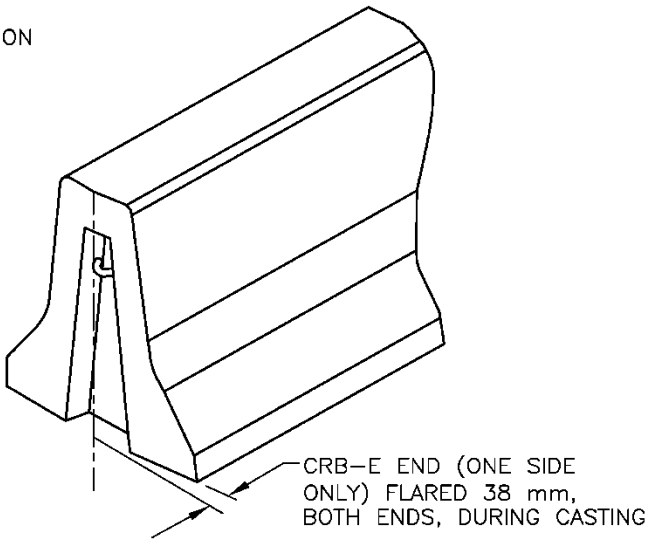


PLAN



ELEVATION

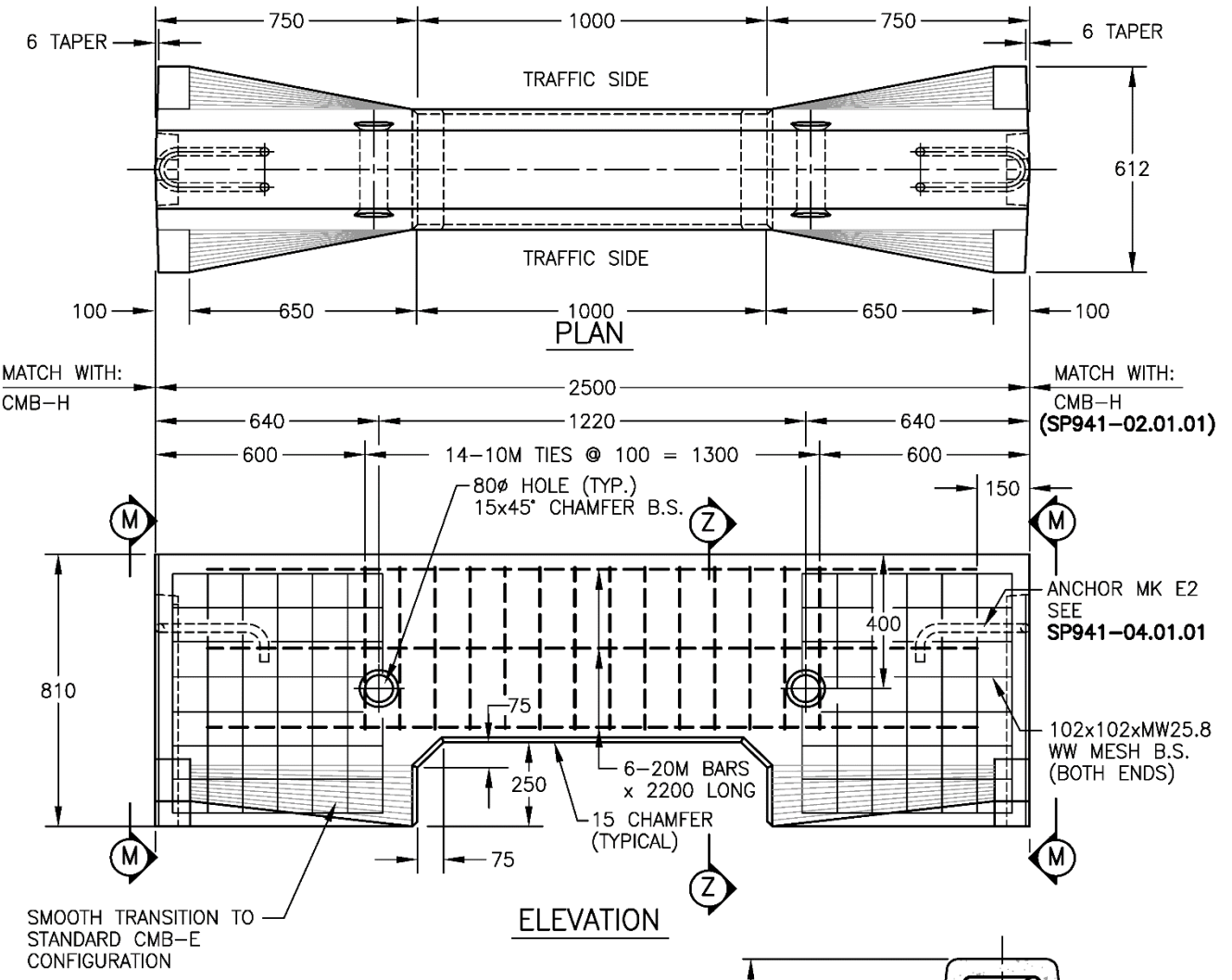
NOTES:
1. THIS DRAWING MUST BE READ WITH CRB DRAWINGS SP941–01.02.01 TO 941–01.02.04 FOR FULL DETAILS OF UNITS.
2. IT IS INTENDED THAT EXISTING CRB MOLDS BE ADAPTED TO DIMENSIONS NOTED ON THIS DRAWING TO PRODUCE UNITS CAPABLE OF BEING PLACED TO A MINIMUM CURVE RADIUS OF APPROXIMATELY 17 m.
3. ALL TOLERANCES AS PER CRB UNIT DRAWINGS.
4. THIS MODIFIED UNIT IS INTENDED TO BE USED ON TIGHT CURVATURE AT INTERSECTIONS ONLY.

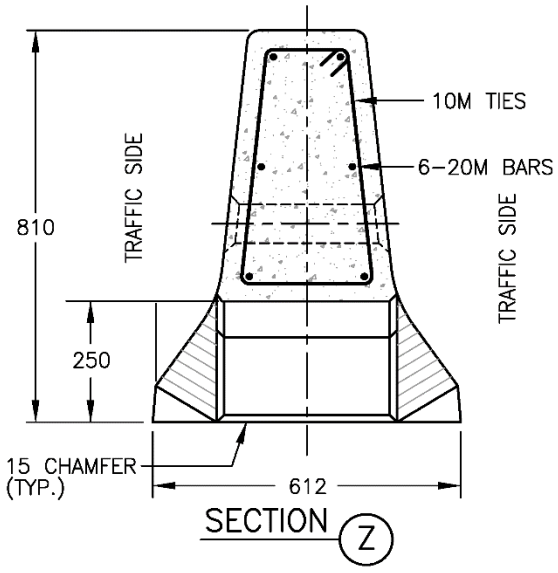


NOT TO SCALE ALL DIMENSIONS ARE IN MILLIMETRES UNLESS OTHERWISE NOTED

PRECAST CONCRETE MEDIAN DRAINAGE BARRIER 810 mm – CMDB–E DETAILS

SP941–02.01.10





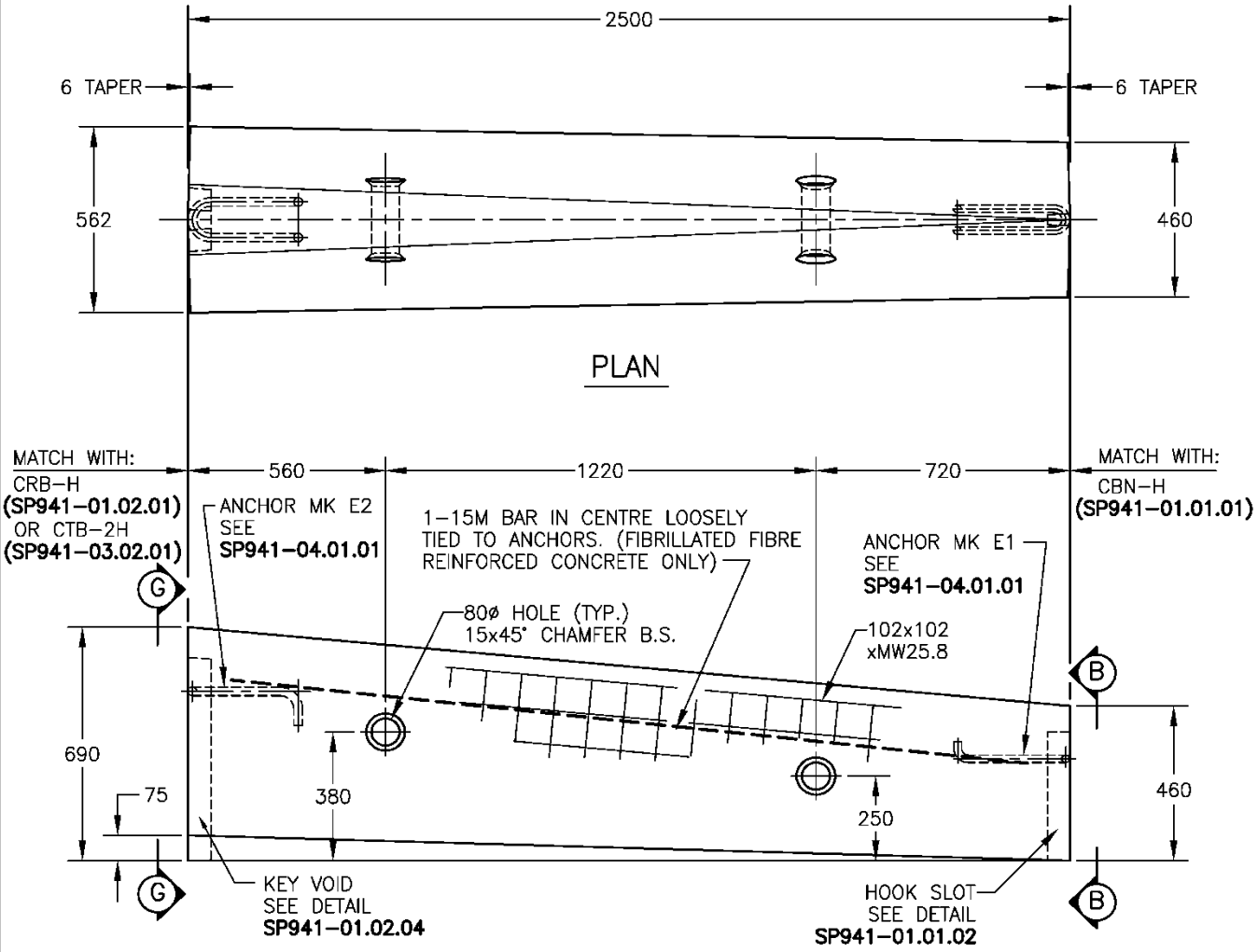
NOTES:

1. FOR VIEW (M) SEE **SP941–02.01.04.**
2. SEE **SP941–02.01.01** FOR DETAILS OF CONNECTING HOOK UNIT.
3. FOR GENERAL NOTES SEE **SP941–01.02.01**
4. FIBRILATED FIBRE STRAND REINFORCED CONCRETE MAY BE SUBSITUTED FOR WELDED WIRE MESH REINFORCED CONCRETE. ADDITIONAL BARS AND TIES WILL STILL APPLY.
5. FOR DETAILS NOT SHOWN SEE **SP941–02.01.02**

NOT TO SCALE ALL DIMENSIONS ARE IN MILLIMETRES UNLESS OTHERWISE NOTED

PRECAST CONCRETE TRANSITION BARRIER 690 mm TO 460 mm – CTB–1E

SP941–03.01.01



NOTES:

1. FOR VIEW (B) SEE **SP941–01.01.02.** FOR VIEW (G) SEE **SP941–01.02.04.**
2. FOR GENERAL NOTES SEE **SP941–01.02.01.**
3. SEE DRAWINGS **SP941–01.01.01** AND **SP941–01.01.02** FOR DETAILS OF 460 mm UNIT AND BULLNOSE. SEE DRAWING **SP941–01.02.01** FOR DETAILS OF 690 mm HOOK UNIT OR **SP941–03.02.01** FOR DETAILS OF 810 TO 690 TRANSITION UNIT.

NOT TO SCALE ALL DIMENSIONS ARE IN MILLIMETRES UNLESS OTHERWISE NOTED